

Pani Simulator

Rishabh Sharma

2024-03-24

Table of contents

Pani Simulator	7
Overview	7
Key Features	9
How to Use This Documentation	10
Setup	11
Prerequisites	11
Cloning the Repository	11
Building the Docker Image	11
Troubleshooting	12
Quickstart	13
Kinematics	17
Overview	17
Coordinate Frames	17
Function Categories	18
Rotation Representation Conversions	18
Quaternion Operations	19
Rate Calculations	19
Navigation Functions	20
Utility Functions	20
Usage Examples	21
Coordinate Transformations	21
Rotation Representations	21
Best Practices	21
Dynamics	23
Overview	23
State Representation	23
Components of the Dynamic Equation	24
Mass Matrix (M)	24
Coriolis and Centripetal Matrix	26
Damping Matrix	27
Restoring Forces	29
Control Forces	29

Vessel ODE and Simulation	30
Helper Methods and Utilities	31
Dimensionalization	31
Hydrodynamic Coefficient Calculation	32
Sensors Simulation	33
Overview	33
Sensor Architecture	33
Sensor Types and Simulation	34
IMU (Inertial Measurement Unit)	34
GPS (Global Positioning System)	35
UWB (Ultra-Wideband Positioning)	36
Encoder Sensors	37
DVL (Doppler Velocity Log)	38
Sensor Configuration	38
Creating Custom Sensors	39
1. Create a new sensor class	39
2. Register your sensor in the factory function	40
3. Create a ROS publisher for your custom sensor	40
4. Configure your custom sensor in the YAML file	41
Best Practices for Sensor Simulation	42
Simulation Inputs	43
Overview	43
Input File Structure	43
Main Configuration File	44
Key Parameters	44
Agent Configuration	45
Vessel-Specific Configuration Files	45
Geometry Configuration (<code>geometry.yml</code>)	45
Inertia Configuration (<code>inertia.yml</code>)	46
Hydrodynamics Configuration (<code>hydrodynamics.yml</code>)	47
Control Surfaces Configuration (<code>control_surfaces.yml</code>)	48
Thruster Configuration (<code>thrusters.yml</code>)	49
Initial Conditions Configuration (<code>initial_conditions.yml</code>)	50
Sensors Configuration (<code>sensors.yml</code>)	51
Guidance Configuration (<code>guidance.yml</code>)	52
NACA Airfoil Data (example: <code>NACA0015.csv</code>)	53
Input File Processing	53
Key Functions	54
Cross-Flow Drag Generation	55
Common Customizations	55
Changing Vessel Dynamics	55

Adding or Modifying Sensors	56
Creating a New Vessel	56
Example: Complete Vessel Configuration	56
ROS2 Architecture	60
Overview	60
Architecture	60
Key Components	62
Main Simulation Script (<code>simulate.py</code>)	62
World Class (<code>class_world.py</code>)	63
World_Node Class (<code>class_world_node_ros2.py</code>)	65
Vessel_Pub_Sub Class (<code>class_vessel_pub_sub_ros2.py</code>)	65
Message Flow	66
Topics and Messages	68
Standard Topics	68
Sensor Topics	69
Sensor Integration	69
Actuator Control	71
Actuator Message Structure (<code>Actuator.msg</code>)	73
World and Vessel Population	74
Running the Simulation	78
Best Practices	78
Vessel Configurator	79
Overview	79
Tutorial	80
Step 1: Load a Vessel Model	80
Step 2: Configure the Vessel	81
Step 3: Placing the vessel center points	85
Step 4: Edit Hydrodynamic Coefficients	86
Step 5: Simulation settings	87
Step 6: Upload the Hydra output file (optional, if you are not using cross-flow drag)	88
Step 7: Generate the YAML files	88
Software Architecture	89
Core Components	90
Simulation Visualization Dashboard	93
Overview	93
Starting the Visualization	95
Main Dashboard	95
Key Features	96
Dashboard Layout	96

Real-time Data Updates	96
3D Visualization	97
Key Features	97
Controls	97
Vessel Model Customization	97
Integration with ROS2	97
Communication Flow	98
Required ROS2 Components	98
Topic Subscriptions	98
Implementation Details	98
Key JavaScript Components	99
Vessel State Representation	99
Customization Options	99
Troubleshooting	99
Connection Issues	99
Missing Vessel Data	100
Browser Compatibility	100
Conclusion	100
Example - Creating a ROS2 Waypoint Tracking Controller	101
Overview	101
Creating the ROS2 Package	101
1. Package Structure	101
2. Update Package Dependencies	102
3. Configure Setup Scripts	102
4. Create Directory Structure	103
Implementing the Controller Logic	104
1. Control Module	104
2. Guidance Control Node Class	106
3. Main Entry Point	110
4. Create a Navigation Node Placeholder	111
Creating the Launch File	111
Building and Launching the Package	113
1. Build the Package	113
2. Launch the System	113
Configuring Waypoints	113
Monitoring System Performance	114
Running the simulator without ROS2	115
Overview	115
Implementation Details	115
The <code>ros_flag</code> Parameter	116

Running the Non-ROS Simulator	117
Simulation Output	117
Implementing Custom Controllers	118
Example: Adding a Simple PID Controller	119
Customizing the Non-ROS Simulator	121
Modifying Simulation Parameters	121
Extending the Simulator	121
Advantages of Non-ROS Mode	123
Limitations	123
Conclusion	123

Pani Simulator

Pani (which means *water* in Hindi) is a high fidelity marine vessel simulator. It is built in a manner such that minimum backend edits are required by the user and the simulator can be both configured and run in an intuitive lightweight GUI application which runs on the web. An attempt to make it as “Microsoft’s Flight Simualtor” but for marine vessels has been made.

The simulator can be used to configure a new marine vessel, simulate sensors such as DVL, IMU, LiDAR and Camera, and control the vessel either via teleop input or custom files for autonomous tasks. The simulator supports multiple agents which is useful for swarm autonomy tasks or to develop better collision avoidance algorithms.

The project has been made open source to allow for further development of this simulator and add additional features to make it more versatile.

Overview

The Simulator is majorly divided into two parts:

- Vessel Configurator
- Vessel Simulation

The Vessel Configurator is what allows you to configure a new marine vessel. You can load a FBX Geometry file, select individual components such as control surfaces, thrusters, sensors and configure their parameters via the interactive GUI. If you like a more geeky experience, you can edit the input files manually as well. There are also options to enter the non-linear hydrodynamics coefficients for the vessel (which we plan to integrate with our in-house system identification algorithm, in future), set vessel geometry parameters and specify initial conditions. This is also where you will setup the simulation parameters such as time, time-step, geofence, GPS datum and physical constants (gravity, density etc). It also supports the output from our in-house hydrodynamics program *HyDRA*, which gives the added mass coefficients and wave hydrodynamics of the vessel. Once you are done with the vessel configuration, the program auto generates the input files required for the simulation to run.

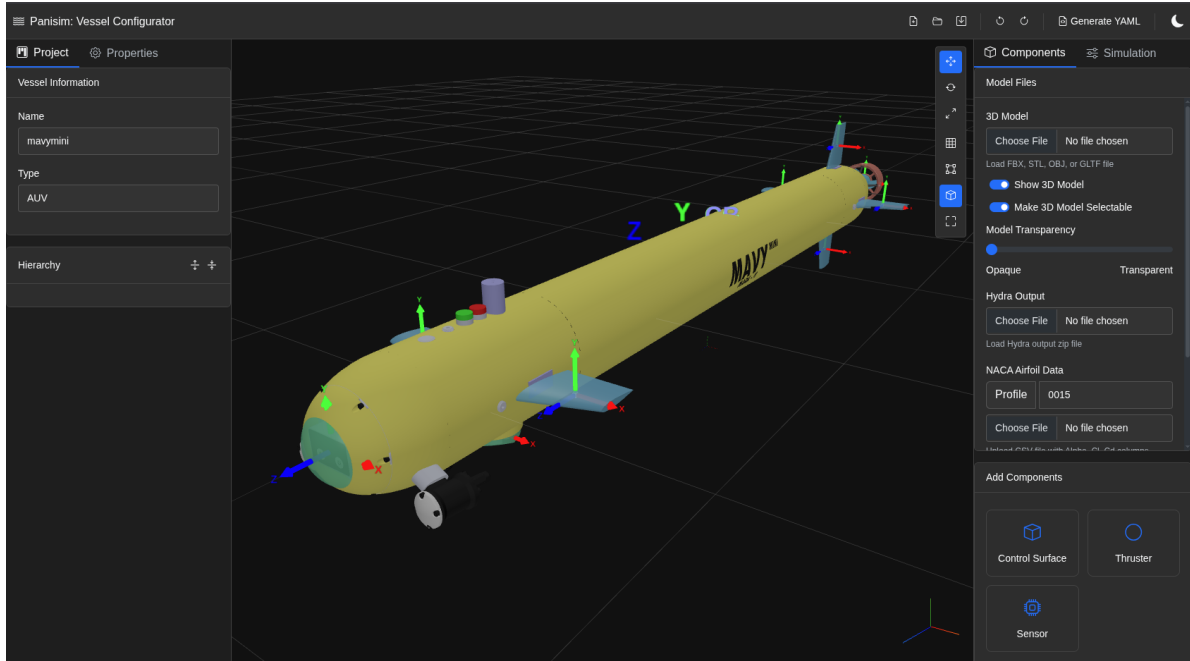


Figure 1: Vessel Configurator

The Vessel Simulation is where the actual simulation takes place. It follows the dynamics as per the parameters you set in the Vessel Configurator. This is where you can also custom code your control and guidance algorithms which will publish the required data in ROS2 to make the vessel run. Currently the sensors are simulated with added noise whose outputs you can see in the ROS2 terminal, or from the GUI. We soon plan to add modelling for the simulator with real world physics.

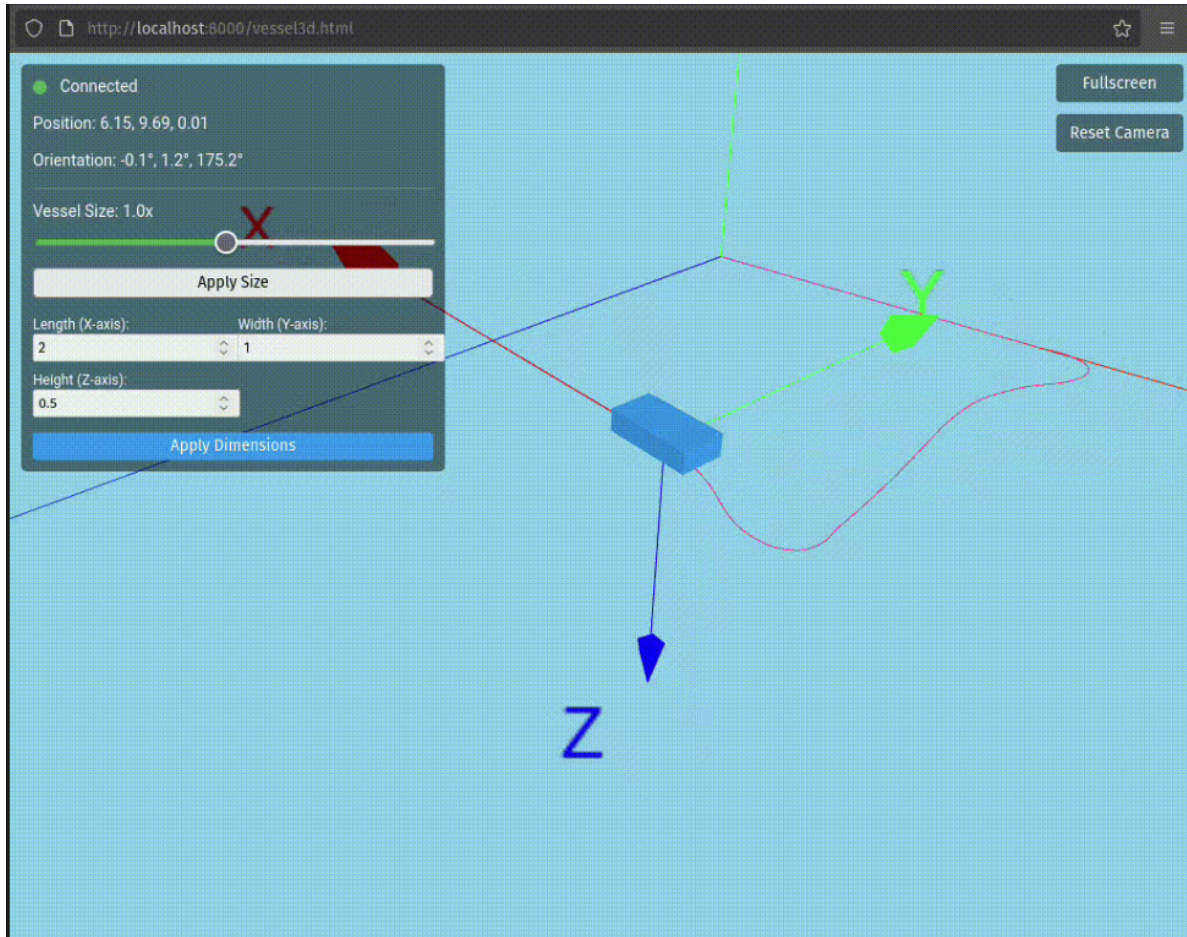


Figure 2: Way-point trackingSimulation Visualization

Key Features

- **Realistic Physics:** Follows dyanmics modelling as defined by Fossen's *Handbook of Marine Craft Hydrodynamics and Motion Control*
- **Modular Architecture:** Easy integration of custom vessel models, sensors, and controllers
- **ROS2 Integration:** Native support for Robot Operating System 2 (ROS2) for distributed robotics applications
- **Web Application:** Lightweight web application for configuring and running the simulation
- **Multiple Agents:** Simulate multiple vessels in the same environment

How to Use This Documentation

Navigate through the sections using the sidebar. The documentation discusses the Setup, theory on the Kinematics and Dynamics used in the simulator, the coordinate systems used for various components (control surfaces, thrusters, sensors and Body centre), the python simulator backend and how to use the Panisim vessel configurator and the Panisim vessel simulator.

Setup

Prerequisites

- Git installed on your system
- Docker installed on your system
- Basic understanding of terminal commands
- Python version 3.10 or higher

Cloning the Repository

1. Open a terminal window
2. Clone the repository using the following command:

```
git clone https://github.com/MarineAutonomy/makara.git
```

3. Navigate to the cloned repository:

```
cd makara
```

4. Switch to the mavymini branch:

```
git checkout mavymini
```

Building the Docker Image

Before running the simulator, you need to build the Docker image:

1. Ensure you're in the root directory of the cloned repository
2. Build the Docker image by executing:

```
./ros2_devdocker.sh
```

3. Wait for the build process to complete (this may take several minutes depending on your internet connection and system performance)

Troubleshooting

- **Docker issues:** Ensure Docker is installed and running on your system
- **Permission issues:** If you encounter permission errors when running scripts, try prefixing the commands with **sudo**

Quickstart

After building the Docker image, you can start the simulator:

1. From the root directory of the repository, run the following script in your terminal:

```
./ros2_simulator.sh
```

2. Once the simulator starts, it will display the ROS2 topics available in the terminal. You should see an output similar to this:

```
=====
ROS2 Simulation Started Successfully!
=====

Available ROS2 Topics:
-----
/parameter_events
/rosout
/sookshma_00/Rudder_1/encoder
/sookshma_00/actuator_cmd
/sookshma_00/imu/data
/sookshma_00/odometry
/sookshma_00/odometry_sim
/sookshma_00/uwb
/sookshma_00/vessel_state
/sookshma_00/vessel_states_ekf
/sookshma_00/waypoints

View Topic Data:
-----

1. Open a new terminal:
   docker exec -it panisim bash

2. Subscribe to a topic:
   ros2 topic echo <topic_name>

Example:
```

```
ros2 topic echo /vessel_0/odometry_sim
```

```
=====
```

To inspect the data published on these topics, follow these steps:

- a. Open a new terminal window.
- b. Enter the Docker container by executing the following command in a new terminal:

```
docker exec -it panisim bash
```

- c. Subscribe to a specific topic using the `ros2 topic echo` command. Replace `<topic_name>` with the actual topic you want to inspect.

```
ros2 topic echo <topic_name>
```

For example, to view the odometry data for `mavymini_00`, you would use:

```
ros2 topic echo /sookshma_00/odometry_sim
```

i Note

The specific topic names may vary depending on the agents activated in your `/inputs/simulation_input.yml` configuration file.

Currently, the example vessel is configured with an initial velocity of 0.5m/s. (You can change this by editing the `/inputs/simulation_input.yml` file.)

A Web-based visualization GUI would have started automatically. You can access it by opening a new browser tab and navigating to `http://localhost:8000`.

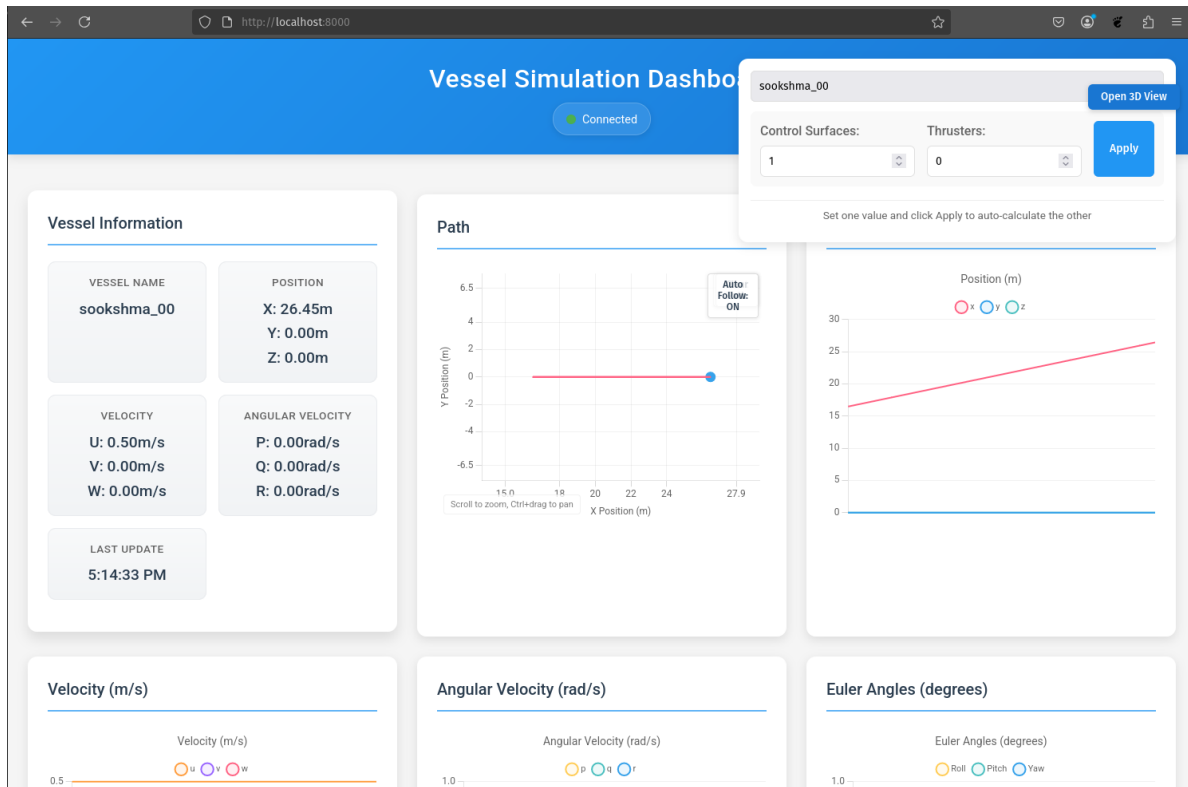


Figure 3: Web-based visualization GUI

This visualization GUI shows you the current state of the vessel, including its position, velocity, orientation and actuator data.

You can also see the 6 DOF of the vessel via the “Open 3D View” button. A new window will appear showing you a box shaped vessel with its position and orientation in the simulation.

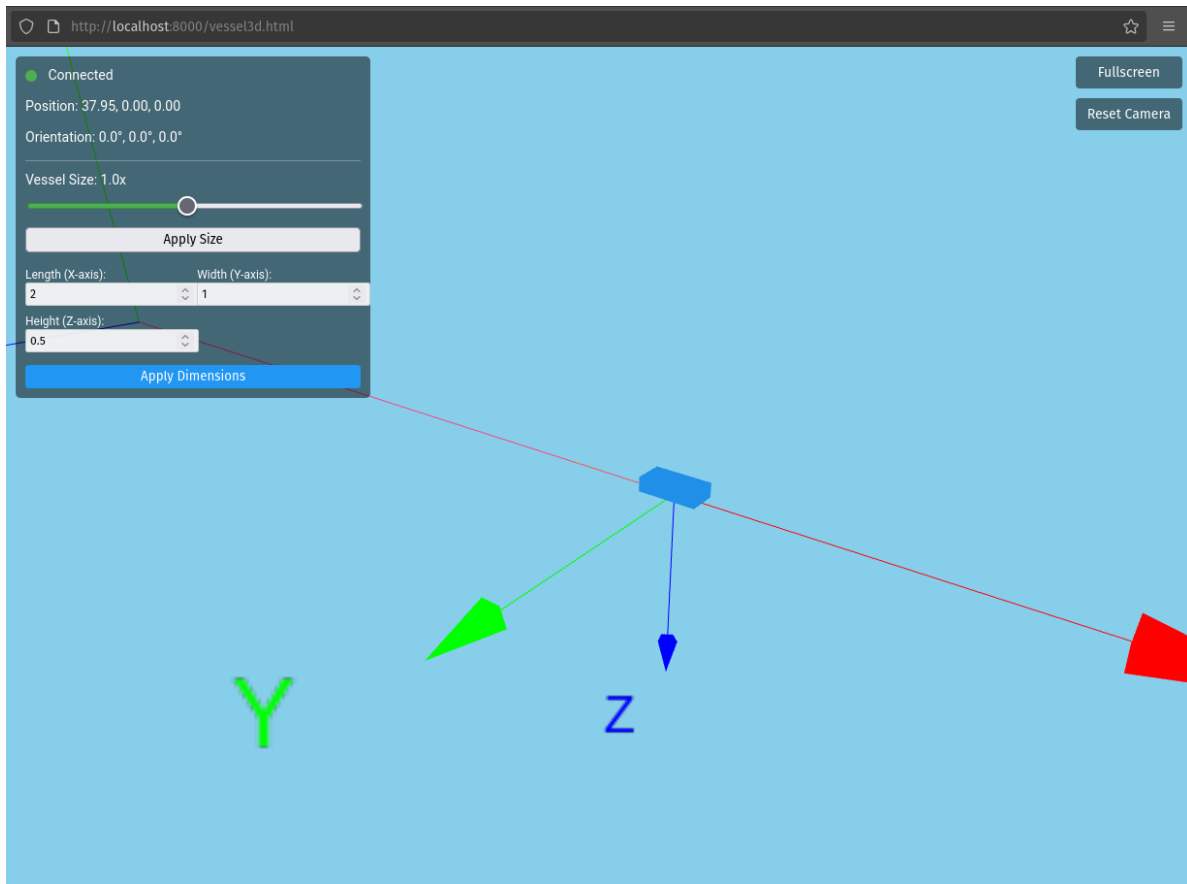


Figure 4: 3D View

And that's it! The simulation is up and running and now you can start any other ROS nodes to command the vessel actuators, perform Extended Kalman Filtering, or anything else you want. We will be covering a waypoint tracking example to show you how you can write your own ROS2 nodes to control your vessel.

Kinematics

Overview

The Kinematics module (`module_kinematics.py`) provides a comprehensive set of mathematical functions for handling coordinate transformations, rotation representations, and kinematic calculations essential for marine vehicle simulation.

Tip

This module serves as the mathematical foundation for all spatial transformations in the simulator.

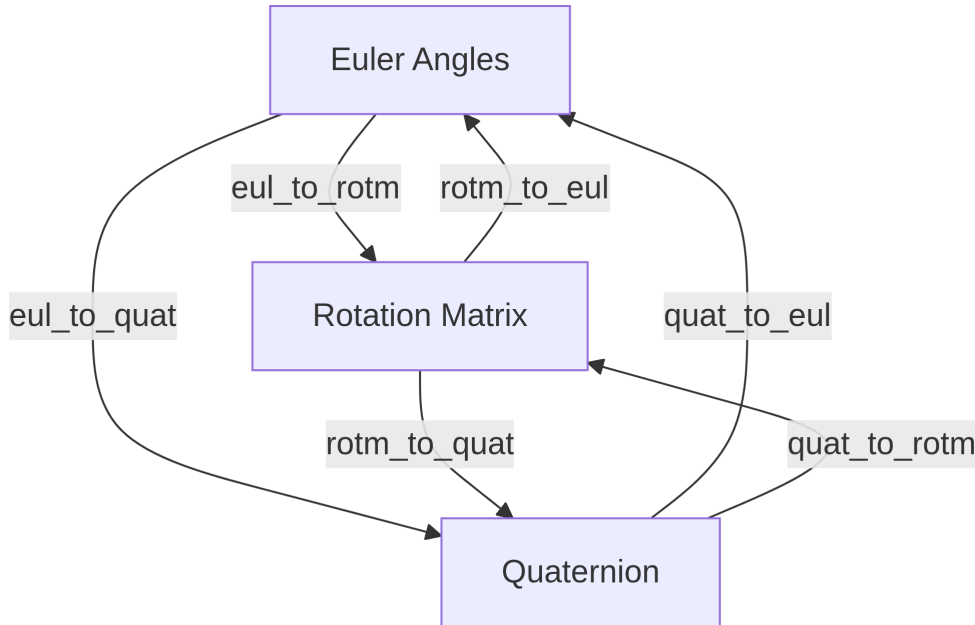
Coordinate Frames

The simulator utilizes several key coordinate frames:

1. **Earth Centered Inertial (ECI) frame $\{i\}$:** An inertial frame with its origin at the Earth's center. It does not rotate with the Earth's rotation.
2. **Earth Centered Earth Fixed (ECEF) frame $\{e\}$:** Rotates with the Earth. Its origin is at the Earth's center. The x-axis passes through the intersection of the prime meridian and the equator, the z-axis aligns with the Earth's rotation axis, and the y-axis completes the right-handed system.
3. **North-East-Down (NED) frame $\{n\}$:** A local tangent frame fixed to a point on the Earth's surface (or a reference point). The x-axis points North, the y-axis points East, and the z-axis points Down, perpendicular to the Earth's ellipsoid surface. This is the primary navigation frame.
4. **Body frame $\{b\}$:** An orthogonal frame fixed to the vehicle. The x-axis points forward, the y-axis points right (starboard), and the z-axis points down.

Function Categories

Rotation Representation Conversions



Function	Description
<code>eul_to_rotm(eul, order='ZYX', deg=False)</code>	Converts Euler angles (roll ϕ , pitch θ , yaw ψ) to a 3x3 rotation matrix that transforms vectors from the Body frame {b} to the NED frame {n}. Assumes ZYX rotation order. Angles can be in radians or degrees.
<code>rotm_to_eul(rotm, order='ZYX', prev_eul=None, deg=False, silent=True)</code>	Converts a 3x3 rotation matrix back to Euler angles (roll ϕ , pitch θ , yaw ψ). Handles potential ambiguities and gimbal lock using previous Euler angles (<code>prev_eul</code>) if provided. Can output in radians or degrees.
<code>eul_to_quat(eul, order='ZYX', deg=False)</code>	Converts Euler angles (roll ϕ , pitch θ , yaw ψ) to a unit quaternion $[q_w, q_x, q_y, q_z]$. Assumes ZYX rotation order. Angles can be in radians or degrees.
<code>quat_to_eul(quat, order='ZYX', deg=False, prev_quat=None, silent=True)</code>	Converts a unit quaternion $[q_w, q_x, q_y, q_z]$ to Euler angles (roll ϕ , pitch θ , yaw ψ). Handles potential ambiguities using <code>prev_quat</code> if provided. Can output in radians or degrees.

Function	Description
<code>quat_to_rotm(quat)</code>	Converts a unit quaternion $[q_w, q_x, q_y, q_z]$ to a 3x3 rotation matrix that transforms vectors from the Body frame {b} to the NED frame {n}.
<code>rotm_to_quat(rotm)</code>	Converts a 3x3 rotation matrix (Body to NED) to a unit quaternion $[q_w, q_x, q_y, q_z]$.

Quaternion Operations

Function	Description
<code>quat_multiply(q1, q2)</code>	Multiplies two quaternions $\mathbf{q}_1 \otimes \mathbf{q}_2$. Order matters: represents rotation $\mathbf{q}_2$ followed by rotation $\mathbf{q}_1$.
<code>quat_conjugate(quat)</code>	Computes the conjugate of a quaternion $\mathbf{q}^* = [q_w, -q_x, -q_y, -q_z]^T$.
<code>rotate_vec_by_quat(vec_a, q_a_b)</code>	Rotates a 3D vector <code>vec_a</code> using the quaternion rotation <code>q_a_b</code> (representing rotation from frame A to frame B) to get the vector in frame B. Computes $\mathbf{v}'_b = \mathbf{q}_{a \rightarrow b} \otimes \mathbf{v}'_a \otimes \mathbf{q}_{a \rightarrow b}^*$.

Rate Calculations

Function	Description
<code>eul_rate_matrix(eul, order='ZYX', deg=False)</code>	Computes the 3x3 transformation matrix $\mathbf{T}(\eta)$ that relates body-frame angular velocity ω^b to Euler angle rates $\dot{\eta}$ via $\dot{\eta} = \mathbf{T}(\eta)\omega^b$. Assumes ZYX order.
<code>quat_rate_matrix(quat)</code>	Computes the 4x3 transformation matrix $\mathbf{E}(\mathbf{q})$ that relates body-frame angular velocity ω^b to quaternion rates $\dot{\mathbf{q}}$ via $\dot{\mathbf{q}} = \frac{1}{2}\mathbf{E}(\mathbf{q})\omega^b$.
<code>eul_rate(eul, w, order='ZYX')</code>	Calculates Euler angle rates $[\dot{\phi}, \dot{\theta}, \dot{\psi}]$ given current Euler angles <code>eul</code> and body-frame angular velocity $\mathbf{w} = [p, q, r]^T$. Uses <code>eul_rate_matrix</code> .
<code>quat_rate(quat, w)</code>	Calculates quaternion rates $[\dot{q}_w, \dot{q}_x, \dot{q}_y, \dot{q}_z]$ given the current quaternion <code>quat</code> and body-frame angular velocity $\mathbf{w} = [p, q, r]^T$. Uses <code>quat_rate_matrix</code> .
<code>deul_dquat(quat)</code>	Computes the 3x4 Jacobian matrix $\frac{\partial \eta}{\partial \mathbf{q}}$, representing the partial derivatives of Euler angles with respect to quaternion components.

Function	Description
<code>dquat_deul(quat)</code>	Computes the 4x3 Jacobian matrix $\frac{\partial \mathbf{q}}{\partial \boldsymbol{\eta}}$, representing the partial derivatives of quaternion components with respect to Euler angles. Requires converting <code>quat</code> to <code>eul</code> internally first.

Navigation Functions

Function	Description
<code>ssa(ang, deg=False)</code>	Converts an angle (in radians or degrees) to its smallest signed angle representation within the range $(-\pi, \pi]$ radians or $(-180, 180]$ degrees.
<code>clip(value, threshold)</code>	Limits the input <code>value</code> to the range <code>[-threshold, threshold]</code> .
<code>ned_to_llh(ned, llh0)</code>	Converts a position vector <code>ned</code> = [North, East, Down] relative to a reference point <code>llh0</code> = [lat0, lon0, h0] into absolute geodetic coordinates <code>llh</code> = [latitude, longitude, height]. Uses WGS84 ellipsoid model.
<code>llh_to_ned(llh, llh0)</code>	Converts an absolute geodetic position <code>llh</code> = [lat, lon, h] into a local NED position vector <code>ned</code> = [North, East, Down] relative to a reference point <code>llh0</code> = [lat0, lon0, h0]. Uses WGS84 ellipsoid model.
<code>generate_waypoints()</code>	Generates a predefined sequence of waypoints as LLH coordinates for a rectangular survey pattern relative to a specific datum location (IITM lake). Returns a numpy array of [lat, lon, height] waypoints.
<code>rotm_ned_to_ecef(llh)</code>	Computes the 3x3 rotation matrix $\mathbf{R}_n^e$ that transforms vectors from the local NED frame (defined at <code>llh</code>) to the ECEF frame.

Utility Functions

Function	Description
<code>Smat(vec)</code>	Creates the 3x3 skew-symmetric matrix $\mathbf{S}(\mathbf{v})$ corresponding to the input 3D vector <code>vec</code> . Used for representing cross products as matrix multiplications ($\mathbf{a} \times \mathbf{b} = \mathbf{S}(\mathbf{a})\mathbf{b}$).

Usage Examples

Coordinate Transformations

```
import numpy as np
from mav_simulator.module_kinematics import eul_to_rotm, llh_to_ned

# Convert Euler angles to rotation matrix
euler_angles = np.array([0.1, 0.2, 0.3]) # [roll, pitch, yaw] in radians
R = eul_to_rotm(euler_angles) # Rotation matrix

# Convert latitude, longitude, height to NED coordinates
reference_point = np.array([60.0, 10.0, 0.0]) # [lat, lon, height]
target_point = np.array([60.001, 10.001, 10.0]) # [lat, lon, height]
ned_coords = llh_to_ned(target_point, reference_point)
```

Rotation Representations

```
from mav_simulator.module_kinematics import eul_to_quat, quat_to_rotm

# Convert from Euler angles to quaternion
euler_angles = np.array([0.1, 0.2, 0.3]) # [roll, pitch, yaw] in radians
quaternion = eul_to_quat(euler_angles)

# Convert from quaternion to rotation matrix
R = quat_to_rotm(quaternion)
```

Best Practices

- Use the `ssa()` function to normalize angles when working with Euler angles

- Be consistent with the rotation order convention throughout your code (the default is ‘ZYX’)
- When transforming between frames, always keep track of the reference frames involved

Dynamics

Overview

The Dynamics modules (`class_vessel.py` and `calculate_hydrodynamics.py`) implement the mathematical foundation for simulating the motion of marine vehicles through water. The simulator uses Fossen's nonlinear 6-DOF equation of motion:

$$M\dot{\nu} + C(\nu)\nu + D(\nu)\nu + g(\eta) = \tau$$

where,

- $M = M_{RB} + M_A$ is the mass matrix, combining rigid body mass M_{RB} and added mass M_A
- $C(\nu) = C_{RB}(\nu) + C_A(\nu)$ is the Coriolis and centripetal matrix
- $D(\nu)$ is the hydrodynamic damping matrix
- $g(\eta)$ is the gravitational and buoyancy force vector
- τ is the control force vector from thrusters and control surfaces
- $\nu = [u, v, w, p, q, r]^T$ is the velocity vector in the body frame
- $\eta = [x, y, z, \phi, \theta, \psi]^T$ is the position and orientation vector

Tip

The dynamics modules serve as the core of the simulation, determining how forces and moments translate into vessel motion through water.

State Representation

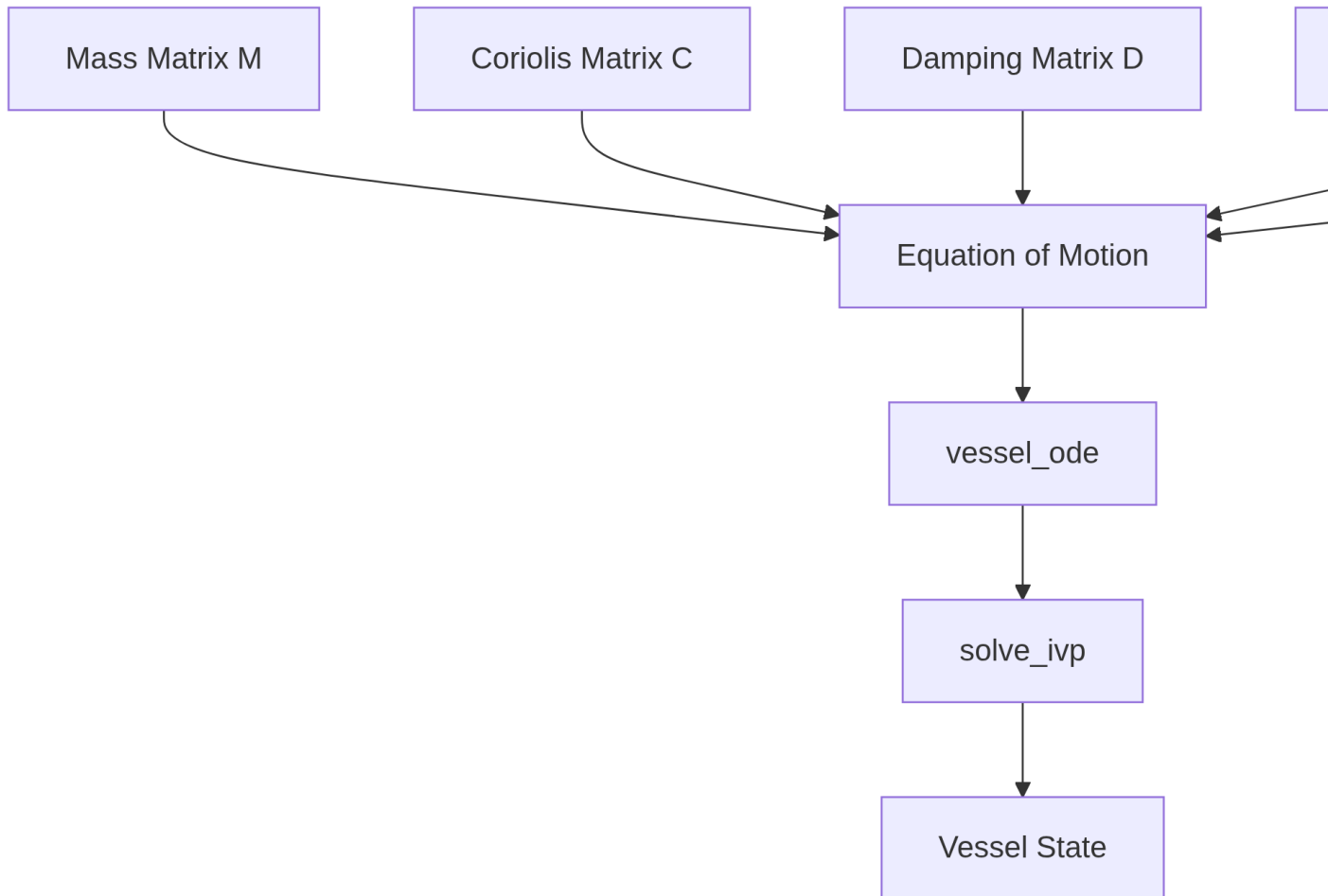
The vessel state is represented as a vector combining velocities, position, orientation, control surface angles, and thruster states:

$$\text{state} = [\nu^T, \eta_{pos}^T, \eta_{rot}^T, \delta^T, n^T]^T$$

Where: - $\nu = [u, v, w, p, q, r]^T$ are the linear and angular velocities in the body frame - $\eta_{pos} = [x, y, z]^T$ is the position in the NED frame - η_{rot} is the orientation, either as Euler angles

$[\phi, \theta, \psi]^T$ or quaternion $[q_0, q_1, q_2, q_3]^T$ - δ is the vector of control surface angles - n is the vector of thruster rotational speeds (RPM)

Components of the Dynamic Equation



Mass Matrix (M)

The mass matrix M combines the rigid-body inertia matrix M_{RB} and the added mass matrix M_A :

$$M = M_{RB} + M_A$$

Rigid-body Mass Matrix (M_{RB})

The rigid-body mass matrix is generated in the `_generate_mass_matrix` method in `calculate_hydrodynamics.py`:

$$M_{RB} = \begin{bmatrix} mI_{3 \times 3} & -mS(r_G) \\ mS(r_G) & I_G \end{bmatrix}$$

Where: - m is the vessel mass - $I_{3 \times 3}$ is the 3×3 identity matrix - r_G is the center of gravity vector relative to the origin of the body frame - $S(r_G)$ is the skew-symmetric matrix of r_G (implemented by `Smat()` in `module_kinematics.py`) - I_G is the inertia tensor calculated from the radii of gyration

```
def _generate_mass_matrix(self, CG, mass, gyration):
    # Calculate gyration tensor about vessel frame
    xprdct2 = np.diag(gyration)**2 - Smat(CG)@Smat(CG)

    # Generate the inertia matrix using radii of gyration
    inertia_matrix = xprdct2 * mass

    # Generate the mass matrix
    mass_matrix = np.zeros((6,6))
    mass_matrix[0:3][:, 0:3] = mass * np.eye(3)
    mass_matrix[3:6][:, 3:6] = inertia_matrix
    mass_matrix[0:3][:, 3:6] = -Smat(CG) * mass
    mass_matrix[3:6][:, 0:3] = Smat(CG) * mass

    return mass_matrix
```

Added Mass Matrix (M_A)

The added mass matrix represents the added inertia due to accelerating the surrounding fluid. It's calculated from hydrodynamic data obtained from [HydRA](#) (You can also enter custom hydrodynamics added mass coefficients (example `Y_vd`) in the `hydrodynamics.yml` input file instead of using HydRA, as discussed in the [Inputs](#) section).

$$M_A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}$$

Function	Description
<code>calculate_added_mass_from_hydra(hydra_file)</code>	Computes the added mass matrix from HydRA data file containing frequency-dependent hydrodynamic coefficients. Extracts zero-frequency added mass for most terms, but computes heave, roll, and pitch terms at their natural frequencies for better accuracy.

```
def calculate_added_mass_from_hydra(self, hydra_file):
    # Load hydrodynamic data
    with open(hydra_file, 'r') as file:
        mdict = json.load(file)

    # Extract arrays from data
    omg = np.array(mdict['w'])      # Frequency array
    AM = np.array(mdict['AM'])      # Added mass array

    # Extract zero-frequency added mass
    A_zero = AM[1, :, :, 0, 0]

    # Calculate natural frequencies and
    # interpolate added mass at those frequencies
    # ...

    # Return properly transformed added mass matrix
    return A
```

i Note

The added mass calculated from HydRA is in ENU frame and so it is converted to NED frame in the `calculate_added_mass_from_hydra` function.

Coriolis and Centripetal Matrix

The Coriolis matrix ($C(\nu)$) accounts for forces arising from motion in a rotating reference frame (Body frame is rotating with respect to NED frame):

$$C(\nu) = C_{RB}(\nu) + C_A(\nu)$$

These matrices are calculated using:

$$C_{RB}(\nu) = \begin{bmatrix} 0_{3 \times 3} & -S(M_{11}v_1 + M_{12}v_2) \\ -S(M_{11}v_1 + M_{12}v_2) & -S(M_{21}v_1 + M_{22}v_2) \end{bmatrix}$$

$$C_A(\nu) = \begin{bmatrix} 0_{3 \times 3} & -S(A_{11}v_1 + A_{12}v_2) \\ -S(A_{11}v_1 + A_{12}v_2) & -S(A_{21}v_1 + A_{22}v_2) \end{bmatrix}$$

Where $v_1 = [u, v, w]^T$ and $v_2 = [p, q, r]^T$ are the linear and angular velocity components of ν .

Function	Description
<code>calculate_coriolis_matrices(vel)</code>	Computes the rigid body and added mass Coriolis-centripetal matrices based on the current velocity state. Returns a tuple of (C_{RB}, C_A) matrices.

Damping Matrix

The damping matrix ($D(\nu)$) represents hydrodynamic resistance forces and is typically modeled as:

$$D(\nu) = D_{linear} + D_{nonlinear}(\nu)$$

In the implementation, damping is handled through hydrodynamic coefficients (entered in the `hydrodynamics.yml` input file) rather than an explicit matrix:

Function	Description
<code>hydrodynamic_forces(vel)</code>	Computes hydrodynamic damping forces by applying coefficients to velocity components. Supports linear, quadratic, and cross-terms.
<code>cross_flow_drag()</code>	Estimates sway-yaw damping coefficients for surface vessels using strip theory and Hoerner's cross-flow drag.
<code>cross_flow_drag_AUV()</code>	Similar to <code>cross_flow_drag()</code> but includes heave-pitch calculations for AUVs.
<code>hoerner()</code>	Implements Hoerner's method for computing 2D drag coefficients based on section beam-to-draft ratio.

The coefficients follow a naming convention that indicates which force/moment they affect and which velocity components are involved (you can enter any custom hydrodynamic coefficients in the `hydrodynamics.yml` input file and it will be handled by the function for force calculations):

- Linear damping: $X_u, Y_v, Z_w, K_p, M_q, N_r$
- Quadratic damping: $X_{u|u}, Y_{v|v}, Z_{w|w}$, etc. (the a indicates absolute value)
- Cross-coupling: X_{vr}, Y_{ur}, N_{uv} , etc.

```
def hydrodynamic_forces(self, vel):
    # Extract velocities
    u, v, w, p, q, r = vel

    # Initialize forces vector
    F = np.zeros(6)

    # Map velocity components to values
    vel_map = {'u': u, 'v': v, 'w': w, 'p': p, 'q': q, 'r': r}

    # Process each hydrodynamic coefficient
    for coeff_name, coeff_value in self.hydrodynamics.items():
        if coeff_value == 0: continue

        parts = coeff_name.split('_')
        force_dir = parts[0]
        if force_dir not in self.force_indices: continue

        # Calculate force contribution
        force = coeff_value
        for component in parts[1:]:
            if component.startswith('a') and len(component) > 1:
                # Absolute value term (e.g., u|u|)
                v_char = component[1:]
                force *= abs(vel_map[v_char])
            elif component in vel_map:
                # Regular term
                force *= vel_map[component]

        # Add to force vector
        F[self.force_indices[force_dir]] += force

    return F
```

Restoring Forces

The restoring forces ($g(\eta)$) include gravitational and buoyancy effects:

$$g(\eta) = \begin{bmatrix} (W - B) \sin \theta \\ -(W - B) \cos \theta \sin \phi \\ -(W - B) \cos \theta \cos \phi \\ -(y_G W - y_B B) \cos \theta \cos \phi + (z_G W - z_B B) \cos \theta \sin \phi \\ (z_G W - z_B B) \sin \theta + (x_G W - x_B B) \cos \theta \cos \phi \\ -(x_G W - x_B B) \cos \theta \sin \phi - (y_G W - y_B B) \sin \theta \end{bmatrix}$$

Where:

- $W = mg$ is the weight
- $B = \rho g \nabla$ is the buoyancy force (with ∇ being the displaced volume)
- $r_G = [x_G, y_G, z_G]^T$ is the center of gravity mentioned in the `geometry.yml` input file
- $r_B = [x_B, y_B, z_B]^T$ is the center of buoyancy mentioned in the `geometry.yml` input file

Function	Description
<code>gravitational_forces(phi, theta)</code>	Computes the gravitational and buoyancy forces and moments as a function of roll and pitch angles.

Control Forces

The control forces vector τ represents the combined effect of control surfaces and thrusters.#### Control Surface Forces

Function	Description
<code>control_forces(delta)</code>	Calculates forces and moments generated by control surfaces (rudders, fins, etc.) based on their deflection angles δ . Supports both hydrodynamic coefficient-based and aerofoil-based calculation methods.

For hydrodynamic coefficient-based calculation: - Forces are calculated as $\tau_i = C_{i\delta} \delta$, where $C_{i\delta}$ is the control coefficient mentioned in the `control_surface_hydrodynamics` in the `control_surfaces.yml` input file

For aerofoil-based calculation:

- Local velocity at the control surface is calculated considering vessel motion (automatically done)
- Angle of attack is determined from flow direction and surface deflection
- Lift and drag forces are calculated using NACA airfoil data (path to the NACA file is mentioned in the `control_surfaces.yml` input file)
- Forces are transformed to the body frame and a generalized force vector is returned

Thruster Forces

Function	Description
<code>thruster_forces(n_prop)</code>	Calculates forces and moments from propellers/thrusters based on their rotational speeds n in RPM. Models thrust using propeller theory with advance ratio and thrust coefficients.

The thrust calculation follows the form:

$$X_{prop} = K_T \rho D^4 |n| n$$

Where:

- K_T is the thrust coefficient (approximated as $K_{T,J=0}(1 - J)$ with J being the advance ratio)
- ρ is the water density
- D is the propeller diameter
- n is the propeller speed in revolutions per second

Vessel ODE and Simulation

The vessel dynamics are implemented as an ordinary differential equation (ODE) in the `vessel_ode(t, state)` method, which computes the state derivatives:

```
def vessel_ode(self, t, state):
    # Extract state components
    vel = state[0:6] # [u, v, w, p, q, r]
    pos = state[6:9] # [x, y, z]
    angles = state[9:attitude_end] # [phi, theta, psi] or quaternion

    # Calculate forces and moments
```

```

F_hyd = self.hydrodynamic_forces(vel)
F_control = self.control_forces(state[control_start:thruster_start])
F_thrust = self.thruster_forces(state[thruster_start:])
F_g = self.gravitational_forces(angles[0], angles[1])

if self.coriolis_flag:
    C_RB, C_A = self.calculate_coriolis_matrices(vel)
    F_C = (C_RB + C_A) @ vel
else:
    F_C = np.zeros(6)

# Total force
F = F_hyd + F_control + F_thrust - F_g - F_C

# Calculate velocity derivatives
M = self.mass_matrix + self.added_mass_matrix # Total mass matrix
state_dot[0:6] = np.linalg.inv(M) @ F

# Calculate position and attitude derivatives
# ...

return state_dot

```

Function	Description
<code>step()</code>	Advances the simulation by one time step by solving the ODE using SciPy's <code>solve_ivp</code> .
<code>simulate()</code>	Runs the full simulation by repeatedly calling <code>step()</code> until the final time.
<code>reset()</code>	Resets the vessel to its initial state.

Helper Methods and Utilities

Dimensionalization

Hydrodynamic coefficients can be provided in non-dimensional form and converted to dimensional form:

Function	Description
<code>_dimensionalize_coefficients(rho, L, U)</code>	Converts non-dimensional hydrodynamic coefficients to dimensional form based on density ρ , length L , and characteristic velocity U (mentioned in the <code>simulation_input.yml</code> file)

Hydrodynamic Coefficient Calculation

Function	Description
<code>calculate_added_mass_from_hydra(hydra_files)</code>	Calculates the added mass matrix from HyDRA data files.
<code>cross_flow_drag()</code>	Estimates sway-yaw hydrodynamic coefficients using strip theory.
<code>cross_flow_drag_AUV()</code>	Estimates both sway-yaw and heave-pitch coefficients for underwater vehicles.
<code>hoerner()</code>	Implements Hoerner's method for 2D section drag coefficient calculation.

Note

Understanding these dynamics components is essential for correctly configuring vessel parameters and interpreting simulation results. The implementation allows for flexible modeling of different vessel types by adjusting the parameters and coefficients.

Sensors Simulation

Overview

The Panisim simulator provides realistic sensor simulations to enable the development and testing of perception, estimation, and control algorithms. The sensor system is designed to be:

- **Modular:** Each sensor is implemented as a separate class
- **Configurable:** Sensors can be customized through YAML configuration files
- **Realistic:** Includes appropriate noise models and physical placement effects
- **Extensible:** New sensor types can be easily added

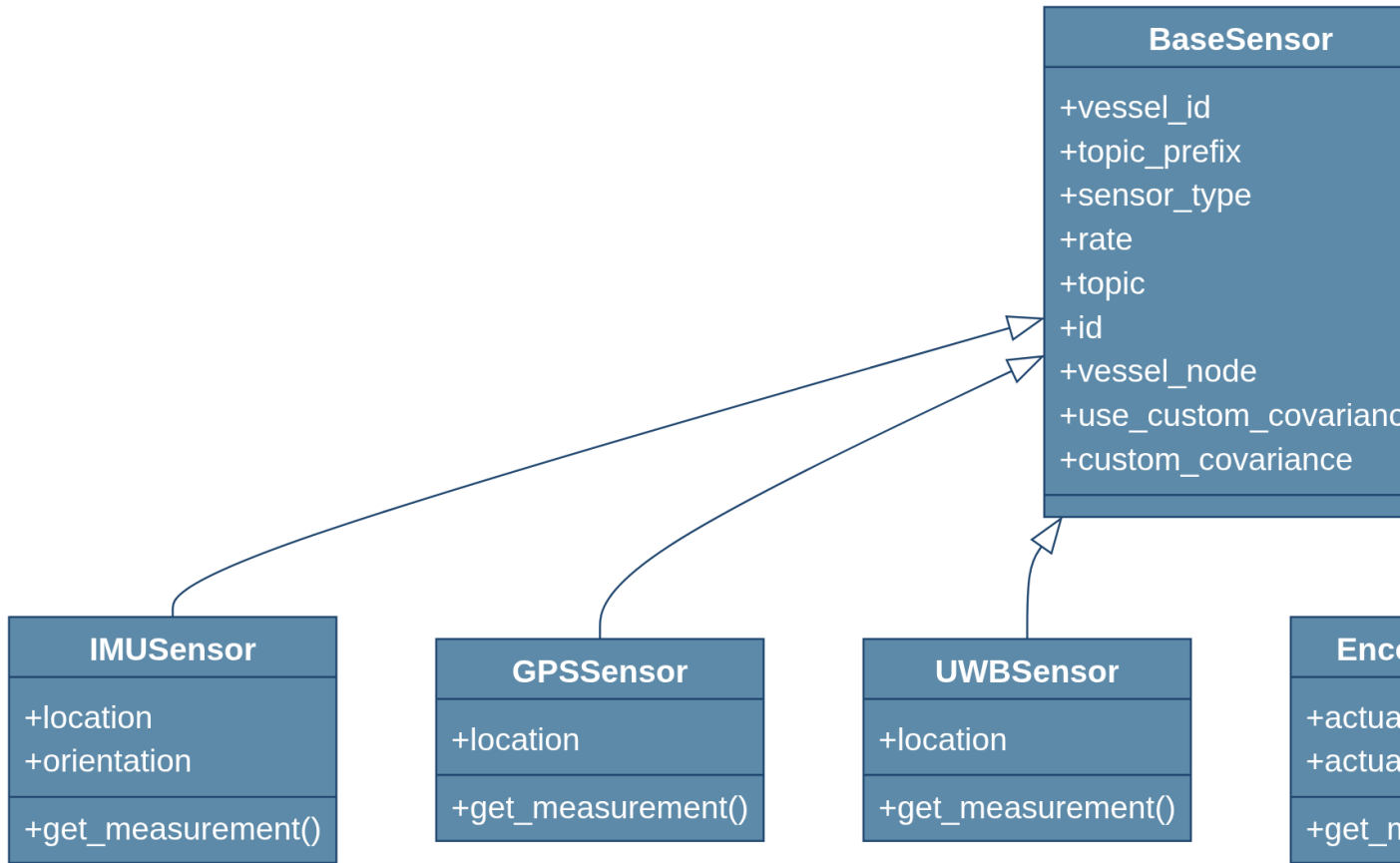
This document explains how sensors are simulated, configured, and how you can create your own custom sensors.

Sensor Architecture

All sensors in the simulator inherit from a common `BaseSensor` class that handles basic functionality such as:

- Sensor identification and topic naming
- Update rate management
- Access to vessel state
- Noise and covariance management

The sensor architecture follows this class hierarchy:



Sensor Types and Simulation

The simulator currently supports the following sensor types:

IMU (Inertial Measurement Unit)

IMU sensors measure: - Orientation (quaternion) - Angular velocity (rad/s) - Linear acceleration (m/s^2)

Simulation details:

The IMU simulation accounts for:

1. Sensor placement relative to the vessel's center of mass
2. Sensor orientation relative to the vessel's body frame
3. Gravitational acceleration effects

4. Centripetal and tangential accelerations due to vessel rotation
5. Realistic noise models for all measurements

Noise model:

- Orientation: Multivariate normal distribution applied to Euler angles
- Angular velocity: Multivariate normal distribution
- Linear acceleration: Multivariate normal distribution

The covariances are defined in the `sensors.yml` file.

```
# Example of how IMU measurements are generated
def get_measurement(self, quat=False):
    state = self.vessel_node.vessel.current_state
    state_der = self.vessel_node.vessel.current_state_der

    # Convert orientation and add noise
    q_sensor = kin.rotm_to_quat(kin.quat_to_rotm(quat) @ kin.quat_to_rotm(orientation_quat))
    q_sensor = kin.eul_to_quat(kin.quat_to_eul(q_sensor) + np.random.multivariate_normal(np.zeros(3), self.lin_acc_cov))

    # Calculate acceleration including all effects
    acc_bcs = state_der[0:3] + np.cross(omg_bcs, v_bcs)
    acc_s_bcs = acc_bcs + np.cross(alpha, self.location) + np.cross(omg_bcs, np.cross(omg_bcs, self.location))
    acc_sensor = kin.quat_to_rotm(orientation_quat).T @ acc_s_bcs

    # Add gravity and noise
    acc_sensor = acc_sensor + kin.quat_to_rotm(q_sensor).T @ np.array([0, 0, -self.vessel_node.vessel.gravity])
    acc_sensor = acc_sensor + np.random.multivariate_normal(np.zeros(3), self.lin_acc_cov)

    # Calculate angular velocity with noise
    omg_sensor = kin.quat_to_rotm(orientation_quat).T @ omg_bcs
    omg_sensor = omg_sensor + np.random.multivariate_normal(np.zeros(3), self.ang_vel_cov)

    return {...} # Return measurements and covariances
```

GPS (Global Positioning System)

GPS sensors measure:

- Latitude (degrees)
- Longitude (degrees)
- Altitude (meters)

Simulation details:

The GPS simulation accounts for:

1. Sensor placement on the vessel
2. Conversion between NED (North-East-Down) and LLH (Latitude-Longitude-Height) coordinates
3. Realistic position noise

Noise model:

- Position: Multivariate normal distribution applied to NED coordinates before conversion to LLH

```
# Example of how GPS measurements are generated
def get_measurement(self, quat=False):
    state = self.vessel_node.vessel.current_state
    llh0 = self.vessel_node.vessel.gps_datum

    # Get position in NED, add sensor offset and noise
    ned = state[6:9] + kin.quat_to_rotm(orientation) @ self.location
    ned = ned + np.random.multivariate_normal(np.zeros(3), self.gps_cov)

    # Convert to latitude-longitude-height
    llh = kin.ned_to_llh(ned, llh0)

    return {...} # Return latitude, longitude, altitude and covariance
```

UWB (Ultra-Wideband Positioning)

UWB sensors measure: - Position in the NED frame (meters)

Simulation details:

The UWB simulation accounts for:

1. Sensor placement on the vessel
2. Position in the NED coordinate frame
3. Realistic position noise (typically higher precision than GPS)

Noise model:

- Position: Multivariate normal distribution applied to NED coordinates

```
# Example of how UWB measurements are generated
def get_measurement(self, quat=False):
    state = self.vessel_node.vessel.current_state
    ned = state[6:9]

    # Get position and add sensor offset and noise
    r_sen = ned + kin.quat_to_rotm(orientation) @ self.location
    r_sen = r_sen + np.random.multivariate_normal(np.zeros(3), self.uwb_cov)

    return {...} # Return position and covariance
```

Encoder Sensors

Encoder sensors measure:

- Actuator position (degrees for control surfaces, RPM for thrusters)

Simulation details:

The encoder simulation accounts for:

1. Specific actuator type (rudder, thruster, etc.)
2. Unit conversion from state vector to appropriate units
3. Actuator-specific noise characteristics

Noise model:

- Actuator value: Normal distribution with specified RMS noise

```
# Example of how Encoder measurements are generated
def get_measurement(self):
    state = self.vessel_node.vessel.current_state

    # Get the specific actuator value from the state vector
    actuator_value_raw = state[self.state_index]

    # Apply unit conversion (e.g., rad to degrees, rad/s to RPM)
    actuator_value_converted = actuator_value_raw * self.unit_conversion

    # Add noise to the converted measurement
    actuator_value_noisy = actuator_value_converted + np.random.normal(0, self.noise_rms)

    return {...} # Return actuator value, name, and covariance
```

DVL (Doppler Velocity Log)

DVL sensors measure:

- Linear velocity in the body frame (m/s)

Simulation details:

The DVL simulation accounts for:

1. Sensor placement and orientation
2. Body-frame velocity measurements
3. Realistic velocity noise

Noise model:

- Velocity: Multivariate normal distribution applied to body-frame velocity

```
# Example of how DVL measurements are generated
def get_measurement(self, quat=False):
    state = self.vessel_node.vessel.current_state

    # Get body-frame velocity and add noise
    v_body = state[0:3]
    v_body_noisy = v_body + np.random.multivariate_normal(np.zeros(3), self.vel_cov)

    return {...} # Return velocity and covariance
```

Sensor Configuration

Sensors are configured through the YAML configuration files. Each sensor definition includes:

- **sensor_type**: The type of sensor (IMU, GPS, UWB, encoder, DVL)
- **sensor_topic**: ROS topic for publishing sensor data
- **sensor_location**: Position of the sensor in the body frame [x, y, z]
- **sensor_orientation**: Orientation of the sensor in the body frame [roll, pitch, yaw]
- **publish_rate**: Update frequency in Hz
- **use_custom_covariance**: Boolean flag to use custom noise parameters
- **custom_covariance**: Custom noise covariance matrices for each measurement

Example configuration for an IMU sensor:

```
sensors:
-
  sensor_type: IMU
  sensor_topic: /vessel_01/imu/data
  sensor_location: [0.0, 0.0, 0.1]
  sensor_orientation: [0.0, 0.0, 0.0]
  publish_rate: 100
  use_custom_covariance: true
  custom_covariance:
    orientation_covariance: [0.001, 0.0, 0.0, 0.0, 0.001, 0.0, 0.0, 0.0, 0.001]
    angular_velocity_covariance: [0.0001, 0.0, 0.0, 0.0, 0.0001, 0.0, 0.0, 0.0, 0.0001]
    linear_acceleration_covariance: [0.01, 0.0, 0.0, 0.0, 0.01, 0.0, 0.0, 0.0, 0.01]
```

Creating Custom Sensors

To create your own custom sensor, follow these steps:

1. Create a new sensor class

Create a new class that inherits from `BaseSensor` and implements the `get_measurement()` method:

```
class CustomSensor(BaseSensor):
    def __init__(self, sensor_config, vessel_id, topic_prefix, vessel_node):
        super().__init__(sensor_config, vessel_id, topic_prefix, vessel_node)
        # Initialize sensor-specific properties
        self.location = np.array(sensor_config['sensor_location'])
        self.custom_param = sensor_config.get('custom_param', default_value)

        # Initialize noise parameters
        self.custom_rms = np.array([0.1, 0.1, 0.1])
        self.custom_cov = np.diag(self.custom_rms ** 2)

        # Override with custom covariance if provided
        if self.use_custom_covariance and self.custom_covariance:
            if 'custom_measurement_covariance' in self.custom_covariance:
                self.custom_cov = np.array(self.custom_covariance['custom_measurement_covariance'])
```

```

def get_measurement(self, quat=False):
    # Access vessel state
    state = self.vessel_node.vessel.current_state

    # Generate realistic measurements based on state
    # Apply appropriate transformations, physics models, etc.

    # Add noise to measurements
    measurement = calculated_value + np.random.multivariate_normal(np.zeros(3), self.custom_cov)

    # Return measurement as a dictionary
    return {
        'custom_measurement': measurement,
        'covariance': self.custom_cov.flatten()
    }

```

2. Register your sensor in the factory function

Modify the `create_sensor` function to include your new sensor type:

```

def create_sensor(sensor_config, vessel_id, topic_prefix, vessel_node):
    """Factory function to create appropriate sensor object based on sensor type"""
    sensor_type = sensor_config['sensor_type']
    sensor_classes = {
        'IMU': IMUSensor,
        'GPS': GPSSensor,
        'UWB': UWBSensor,
        'encoder': EncoderSensor,
        'DVL': DVLSensor,
        'CustomSensor': CustomSensor # Add your custom sensor here
    }

    if sensor_type not in sensor_classes:
        raise ValueError(f"Unknown sensor type: {sensor_type}")

    return sensor_classes[sensor_type](sensor_config, vessel_id, topic_prefix, vessel_node)

```

3. Create a ROS publisher for your custom sensor

In your vessel's ROS node, add a publisher for your custom sensor:


```

# In the vessel ROS node initialization
if sensor.sensor_type == 'CustomSensor':
    # Create a publisher for your custom sensor
    # Choose an appropriate message type or create a custom one
    from your_package.msg import CustomSensorMsg

    publisher = self.create_publisher(
        CustomSensorMsg,
        f'{self.topic_prefix}/{sensor.topic}',
        10
    )

    # Add the publisher to the sensor publishers dictionary
    self.sensor_publishers[sensor] = publisher

# In the vessel ROS node update method
if sensor.sensor_type == 'CustomSensor':
    # Create a message for your custom sensor
    msg = CustomSensorMsg()

    # Fill the message with sensor data
    measurement = sensor.get_measurement()
    msg.data = measurement['custom_measurement']
    msg.covariance = measurement['covariance']

    # Add a timestamp
    msg.header.stamp = self.get_clock().now().to_msg()

    # Publish the message
    self.sensor_publishers[sensor].publish(msg)

```

4. Configure your custom sensor in the YAML file

Add your custom sensor to the sensors configuration file:

```

sensors:
-
    sensor_type: CustomSensor
    sensor_topic: /vessel_01/custom_sensor
    sensor_location: [0.1, 0.0, 0.2]
    sensor_orientation: [0.0, 0.0, 0.0]

```

```
publish_rate: 50
custom_param: 42.0
use_custom_covariance: true
custom_covariance:
  custom_measurement_covariance: [0.01, 0.0, 0.0, 0.0, 0.01, 0.0, 0.0, 0.0, 0.01]
```

Best Practices for Sensor Simulation

When working with sensors in the simulator, follow these guidelines:

1. Physical realism:

- Place sensors at realistic positions on the vessel
- Use realistic noise parameters based on actual sensor specifications
- Consider environmental effects if appropriate (e.g., GPS degradation underwater)

2. Computational efficiency:

- Set appropriate update rates based on real-world sensors
- Use optimized noise generation methods for high-frequency sensors

3. Sensor fusion testing:

- Configure multiple sensor types to test fusion algorithms
- Vary noise parameters to test robustness of estimation algorithms

4. Custom sensors:

- Follow the class structure of existing sensors
- Document physical models and assumptions clearly
- Use appropriate coordinate transformations

5. Topic naming:

- Use consistent naming conventions for sensors
- Include vessel ID in topic names for multi-vessel simulations

Simulation Inputs

Overview

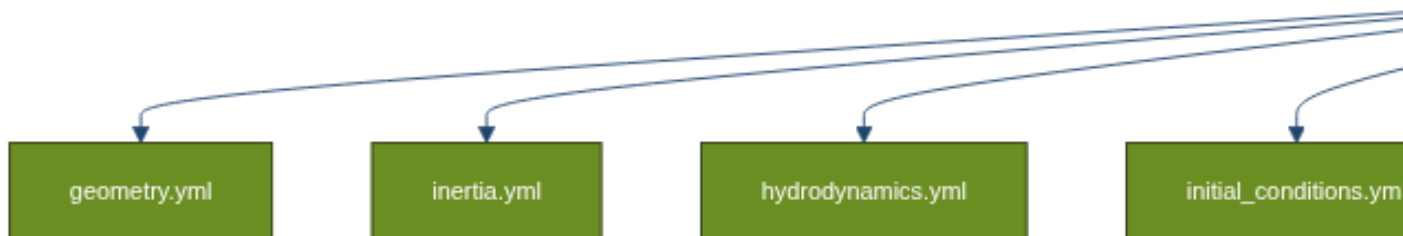
The Panisim simulator uses a structured set of YAML configuration files to define all aspects of the simulation environment, vessel properties, sensor configurations and guidance. This modular approach allows for easy customization and extension of simulation scenarios without modifying the core simulation code. The input files are auto generated via the interactive Web GUI, but can also be modified manually.

💡 Tip

All input files follow the YAML format, which is human-readable and easy to edit.

Input File Structure

The input file system follows a hierarchical structure (zoom in to see the chart more clearly).



The top-level `simulation_input.yml` file defines global simulation parameters and references vessel-specific configuration files stored in separate folders (one for each vessel type).

Main Configuration File

The `simulation_input.yml` file is the entry point for the simulator configuration. It specifies global parameters and the list of agents (vessels) to include in the simulation.

Example structure:

```
sim_time: 100.0          # Total simulation time in seconds
time_step: 0.01          # Simulation time step in seconds
density: 1025.0          # Water density (kg/m³)
gravity: 9.80665         # Gravitational acceleration (m/s²)
world_size: [1000, 1000, 100] # Simulation world dimensions [x, y, z] in meters
gps_datum: [13.06, 80.28, 0] # GPS reference point [latitude, longitude, height]
nagents: 1              # Number of agents in the simulation
geofence: []            # Optional geofencing boundaries

agents:
  - name: "sookshma"      # Vessel name
    type: "usv"           # Vessel type (usv, auv, etc.)
    active_dof: [1, 1, 0, 0, 0, 1] # Active degrees of freedom [u, v, w, p, q, r]
    U: 0.0                # Initial forward speed
    maintain_speed: false # Whether to maintain constant forward speed

    # Paths to configuration files (relative or absolute)
    # {name} will be replaced with the vessel name
    geometry: "inputs/{name}/geometry.yml"
    inertia: "inputs/{name}/inertia.yml"
    hydrodynamics: "inputs/{name}/hydrodynamics.yml"
    control_surfaces: "inputs/{name}/control_surfaces.yml"
    thrusters: "inputs/{name}/thrusters.yml"
    initial_conditions: "inputs/{name}/initial_conditions.yml"
    sensors: "inputs/{name}/sensors.yml"
    guidance: "inputs/{name}/guidance.yml"
    control: "inputs/{name}/control.yml"
```

Key Parameters

Parameter	Description	Units
<code>sim_time</code>	Total simulation duration	seconds
<code>time_step</code>	Simulation time step size	seconds
<code>density</code>	Water density	kg/m ³
<code>gravity</code>	Gravitational acceleration	m/s ²
<code>world_size</code>	Simulation world dimensions [x, y, z]	meters
<code>gps_datum</code>	Reference point for GPS coordinates [lat, lon, height]	degrees, degrees, meters
<code>nagents</code>	Number of agents (vessels) in the simulation	integer
<code>geofence</code>	Optional boundaries for the simulation area	meters

Agent Configuration

Each agent (vessel) in the `agents` list requires:

- **name**: Identifier for the vessel, used for file path resolution
- **type**: Type of vessel (“usv”, “auv”, etc.)
- **active_dof**: Array of 6 binary values indicating which degrees of freedom are active [surge, sway, heave, roll, pitch, yaw]
- **U**: Initial/desired forward speed
- **maintain_speed**: Boolean flag to maintain constant forward speed

File paths can be absolute or relative (although we would recommend using the same folder structure as the default). The special placeholder `{name}` will be replaced with the vessel name, allowing for a standardized folder structure.

Vessel-Specific Configuration Files

Each vessel has its own set of configuration files that define its physical properties, sensors, and control elements. This is defined in a folder named after the vessel.

Geometry Configuration (`geometry.yml`)

The geometry file defines the vessel’s physical dimensions and coordinate frames.

Example from sookshma:

```
# Dimensions of the box enclosing the vessel
length: 1
breadth: 0.24
depth: 0.5
Fn: 0.16
gyration: [0.096, 0.25, 0.25]
C0:
  position: [0.0, 0.0, 0.0] # Relative to the origin of the 3d software
  orientation: [0.0, 0.0, 0.0]
CG:
  position: [2.89415958e-02, 0.0, 1.62558889e-01]
CB:
  position: [2.89415958e-02, 0.0, 1.36000000e-01]

geometry_file: /workspaces/mavlab/inputs/sookshma/HydRA/input/sookshma.gdf
```

Key Parameters

Parameter	Description	Units
length	Overall length	meters
breadth	Overall width	meters
depth	Overall depth	meters
Fn	Froude number	dimensionless
gyration	Gyration radii [x, y, z]	meters
C0	Coordinate origin position and orientation	meters, radians
CG	Center of gravity position and orientation	meters, radians
CB	Center of buoyancy position and orientation	meters, radians
geometry_file	Path to 3D geometry file	string

Inertia Configuration (inertia.yml)

The inertia file defines the vessel's mass properties.

Example from sookshma:

```
mass: 7.65 # in kg
buoyancy_mass: 7.65 # in kg
inertia_matrix: None # in kg*m^2
added_mass_matrix: None # in kg*m^2/s
```

Key Parameters

Parameter	Description	Units
mass	Vessel mass	kg
buoyancy_mass	Mass of displaced fluid (for neutral buoyancy, equal to mass)	kg
inertia_matrix	Optional explicit inertia matrix needs to be 6x6 (example: $\begin{bmatrix} 1, & 0, & 0, & 0, & 0, & 0 \\ 0, & 1, & 0, & 0, & 0, & 0 \\ 0, & 0, & 1, & 0, & 0, & 0 \\ 0, & 0, & 0, & 1, & 0, & 0 \\ 0, & 0, & 0, & 0, & 1, & 0 \\ 0, & 0, & 0, & 0, & 0, & 1 \end{bmatrix}$)	$\text{kg} \cdot \text{m}^2$
added_mass_matrix	Optional explicit added mass matrix needs to be 6x6	kg, $\text{kg} \cdot \text{m}^2$, $\text{kg} \cdot \text{m}$

If the inertia matrix is not provided, it is calculated using the gyration radii and the mass via the `_generate_mass_matrix()` function in the `calculate_hydrodynamics.py` module. If the added mass matrix is not provided, it will be calculated either using the hydrodynamic derivatives or using the hydrodynamic coefficients from the HydRA model.

Hydrodynamics Configuration (hydrodynamics.yml)

The hydrodynamics file contains coefficients for damping forces and added mass.

Example from sookshma:

```
hydra_file: /workspaces/mavlab/inputs/sookshma/HydRA/matlab/sookshma_hydra.json
dim_flag: false
cross_flow_drag: false
coriolis_flag: false # For linear dynamic model

# Surge hydrodynamic derivatives
X_u: -0.01

# Sway hydrodynamic derivatives
Y_v: -0.010659050686665396
Y_r: 0.0017799664706713543

# Yaw hydrodynamic derivatives
N_v: -0.0006242757399292063
N_r: -0.001889456681929024
```

Key Parameters

Parameter	Description	Units
hydra_file	Path to HyDRA data file	string
dim_flag	Whether coefficients are dimensional (true) or non-dimensional (false)	boolean
cross_flow_drag	Whether to use cross-flow drag calculation	boolean
coriolis_flag	Whether to include Coriolis forces	boolean

You can enter any combination of the hydrodynamic coefficients, and it'll be handled by the simulator. They should be separated by underscores. For example, `X_u_u` or `M_r_r`.

Control Surfaces Configuration (`control_surfaces.yml`)

This file defines the vessel's control surfaces such as rudders, fins, or foils.

Example from sookshma:

```
naca_number: &naca None
naca_file: &naca_file None
delta_max: &dmax 35.0
deltad_max: &ddmax 1.0
area: &area 0.1

control_surfaces:
-
  control_surface_type: Rudder
  control_surface_id: 1
  control_surface_location: [0.0, 0.0, 0.0] # With respect to BODY frame
  control_surface_orientation: [0.0, 0.0, 0.0] # With respect to BODY frame
  control_surface_area: *area
  control_surface_NACA: *naca
  control_surface_T: 0.1
  control_surface_delta_max: *dmax
  control_surface_deltad_max: *ddmax
  control_surface_hydrodynamics:
    Y_delta: 0.02100751285923063
    N_delta: -0.006813248905845932
    X_delta: 0.0
```


Key Parameters

Parameter	Description	Units
naca_number	NACA airfoil designation	string
naca_file	Path to NACA airfoil data file	string
delta_max	Maximum deflection angle	degrees
deltad_max	Maximum deflection rate	degrees/s
area	Surface area	m ²
control_surface_type	Type of control surface (e.g., Rudder, Fin)	string
control_surface_id	Unique identifier	integer
control_surface_location	[x, y, z] position	meters
control_surface_orientation	[roll, pitch, yaw] orientation	radians
control_surface_T	Actuation time constant	seconds
control_surface_hydrodynamics	Force coefficients for the control surface	various

If `control_surface_hydrodynamics` is provided, the control surface location, orientation and naca file will be ignored. And the generalized force vector due to control surface will be calculated using the hydrodynamic coefficients.

Thrust Configuration (`thrusters.yml`)

This file defines the vessel's propulsion systems.

Example from sookshma (commented out):

```
thrusters:
-
  thruster_name: back
  thruster_id: 1
  thruster_location: [0.0, 0.0, 0.0] # With respect to BODY frame
  thruster_orientation: [0.0, 0.0, 0.0] # With respect to BODY frame
  T_prop: 1.0
  D_prop: 0.1
  tp: 0.1
  KT_at_J0: 0.0
  n_max: 2668 # in RPM
  J_vs_KT: None # File path to J_vs_KT.csv, Not being used currently
```

Key Parameters

Parameter	Description	Units
thruster_name	Descriptive name	string
thruster_id	Unique identifier	integer
thruster_location	[x, y, z] position	meters
thruster_orientation	[roll, pitch, yaw] orientation	radians
T_prop	Propeller thrust coefficient	dimensionless
D_prop	Propeller diameter	meters
tp	Thrust deduction coefficient	dimensionless
KT_at_J0	Thrust coefficient at zero advance ratio	dimensionless
n_max	Maximum RPM	RPM
J_vs_KT	Path to file with advance ratio vs thrust coefficient data	string

Initial Conditions Configuration (initial_conditions.yml)

This file defines the vessel's initial state for the simulation.

Example from sookshma:

```
start_location: [0, 0, 0]
start_orientation: [0, 0, 0]
start_velocity: [0.5, 0, 0, 0, 0, 0]
use_quaternion: false
```

Key Parameters

Parameter	Description	Units
start_location	Initial [x, y, z] position	meters
start_orientation	Initial [roll, pitch, yaw] orientation	radians
start_velocity	Initial [u, v, w, p, q, r] velocity	m/s, rad/s
use_quaternion	Whether to use quaternions for orientation	boolean

Sensors Configuration (sensors.yml)

This file defines the sensors attached to the vessel.

Example from sookshma:

```
# Custom covariance flag is applicable only when using GNC

sensors:
-
  sensor_type: IMU
  sensor_topic: /sookshma_00/imu/data
  sensor_location: [0.0, 0.0, 0.0]
  sensor_orientation: [0.0, 0.0, 0.0]
  publish_rate: 50
  use_custom_covariance: true
  custom_covariance:
    orientation_covariance: [4.97116638e-07, 1.92100383e-07, -5.37921803e-06, 1.92100383e-07, -5.37921803e-06, 1.92100383e-07,
    linear_acceleration_covariance: [0.01973958, -0.01976063, 0.02346221, -0.01976063, 0.02346221, -0.01976063,
    angular_velocity_covariance: [5.28022053e-05, 4.08840955e-05, -1.15368805e-05, 4.08840955e-05, -1.15368805e-05, 5.28022053e-05]

-
  sensor_type: UWB
  sensor_topic: /sookshma_00/uwb
  sensor_location: [0.0, 0.0, 0.0]
  sensor_orientation: [0.0, 0.0, 0.0]
  publish_rate: 1
  use_custom_covariance: true
  custom_covariance:
    position_covariance: [0.04533883, -0.05014115, 0.0, -0.05014115, 0.05869406, 0.0, 0.0,
    orientation_covariance: [0.0001, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0001]

-
  sensor_type: encoder
  actuator_type: Rudder
  actuator_id: 1
  sensor_topic: /sookshma_00/Rudder_1/encoder
  publish_rate: 10
```

Key Parameters

Parameter	Description	Units
<code>sensor_type</code>	Type of sensor (IMU, GPS, UWB, DVL, encoder, etc.)	string
<code>sensor_topic</code>	ROS topic for publishing sensor data	string
<code>sensor_location</code>	[x, y, z] position	meters
<code>sensor_orientation</code>	[roll, pitch, yaw] orientation	radians
<code>publish_rate</code>	Data publication frequency	Hz
<code>use_custom_covariance</code>	Whether to use custom noise covariance	boolean
<code>custom_covariance</code>	Sensor-specific covariance matrices	various
<code>actuator_type</code>	For encoders, the type of actuator	string
<code>actuator_id</code>	For encoders, the ID of the actuator	integer

Guidance Configuration (`guidance.yml`)

This file defines waypoints and guidance parameters for autonomous navigation as used in the PID Waypoint Tracking Example.

Example from `sookshma`:

```
waypoints:
- [0, 0, 0]
- [-4, -10, 0]
- [8, -5, 0]
- [9, 9, 0]
- [0, 0, 0]
waypoints_type: XYZ
```

Key Parameters

Parameter	Description	Units
<code>waypoints</code>	List of waypoint coordinates	meters
<code>waypoints_type</code>	Coordinate system for waypoints (XYZ, LLA, NED)	string
<code>lookahead_distance</code>	Optional: Distance for lookahead-based guidance	meters
<code>acceptance_radius</code>	Optional: Radius to consider waypoint reached	meters

Parameter	Description	Units
<code>guidance_type</code>	Optional: Type of guidance algorithm	string

NACA Airfoil Data (example: `NACA0015.csv`)

This file contains aerodynamic/hydrodynamic coefficients for the NACA0015 airfoil profile, used for modeling control surfaces like rudders. The file includes:

- **Alpha:** Angle of attack in degrees
- **Cl:** Lift coefficient
- **Cd:** Drag coefficient
- **Cdp:** Pressure drag coefficient
- **Cm:** Pitching moment coefficient
- **Top_Xtr:** Top surface transition location
- **Bot_Xtr:** Bottom surface transition location

Sample excerpt:

```
Alpha,Cl,Cd,Cdp,Cm,Top_Xtr,Bot_Xtr
-19.750,-1.2970,0.09858,0.09490,-0.0142,1.0000,0.0253
-19.500,-1.3072,0.09342,0.08964,-0.0166,1.0000,0.0255
...
0.000,0.0000,0.00632,0.00147,0.0000,0.6107,0.6107
...
19.750,1.2970,0.09858,0.09490,0.0142,0.0253,1.0000
```

This data is used to calculate the forces and moments generated by control surfaces at different angles of attack. This file is obtained from [Airfoil Tools website](#).

Input File Processing

The `read_input.py` module is responsible for loading and processing all input files. The main function is `read_input()`, which:

1. Loads the main simulation configuration file
2. Extracts global simulation parameters
3. Processes each vessel's configuration files
4. Transforms component positions and orientations to be relative to the vessel's coordinate origin (CO)
5. Returns the processed simulation parameters and agent configurations

Key Functions

`read_input(input_file)`

The main entry point that reads the entire configuration:

```
def read_input(input_file: str = None) -> Tuple[Dict, List[Dict]]:
    """Read vessel parameters and configuration from input files.

    Args:
        input_file: Path to the main simulation input file. If None, uses default.

    Returns:
        Tuple containing:
            sim_params (Dict): Global simulation parameters
            agents (List[Dict]): List of agents, each containing vessel config and hydrodynamic parameters
    """
```

`load_yaml(file_path)`

Loads a YAML file and returns its contents:

```
def load_yaml(file_path: str) -> Dict:
    """Load a YAML file and return its contents."""
    with open(file_path, 'r') as file:
        return yaml.safe_load(file)
```

`resolve_path(base_path, file_path, name)`

Resolves file paths, replacing {name} with the actual vessel name:

```
def resolve_path(base_path: str, file_path: str, name: str) -> str:
    """Resolve a file path, replacing {name} with the actual vessel name."""
    resolved = file_path.replace('{name}', name)
    if not os.path.isabs(resolved):
        resolved = os.path.join(base_path, resolved)
    return resolved
```

```
transform_to_co_frame(vessel_config)
```

Transforms all component positions and orientations to be relative to the vessel's coordinate origin (CO):

```
def transform_to_co_frame(vessel_config: Dict) -> Dict:
    """Transform all component positions and orientations to be relative to the CO frame."""
    # Implementation details...
```

Cross-Flow Drag Generation

The `read_input` function also initializes a `CrossFlowGenerator` to calculate cross-flow drag coefficients:

```
cross_flow_generator = CrossFlowGenerator(gdf_file, hydra_file, hydrodynamics_file,
                                          initial_conditions_file, agent['type'])
cross_flow_generator.update_yaml_file()
```

This automatically computes and updates the hydrodynamic coefficients for sway-yaw (and heave-pitch for AUVs) using strip theory and Hoerner's cross-flow drag formulation. This happens if the `cross_flow_drag` flag is set to `true` in the `hydrodynamics.yml` file.

Common Customizations

Here are some common customizations you might want to make to the input files:

Changing Vessel Dynamics

To adjust a vessel's dynamic behavior:

1. Modify hydrodynamic coefficients in `hydrodynamics.yml`
2. Adjust mass and inertia properties in `inertia.yml`
3. Change the center of gravity or buoyancy in `geometry.yml`

Adding or Modifying Sensors

To add a new sensor:

1. Add a new sensor configuration to `sensors.yml` with appropriate parameters
2. Set a realistic update rate and noise characteristics
3. Position the sensor at an appropriate location on the vessel

Creating a New Vessel

To create a new vessel type:

1. Create a new folder with the vessel name
2. Copy and modify the configuration files from an existing vessel
3. Adjust all physical parameters to match the new vessel's characteristics
4. Add the new vessel to the `agents` list in `simulation_input.yml`

Example: Complete Vessel Configuration

Here's an example of a complete vessel configuration for a surface vessel:

1. `simulation_input.yml`:

```
sim_time: 100.0
time_step: 0.01
density: 1025.0
gravity: 9.80665
world_size: [1000, 1000, 100]
gps_datum: [13.06, 80.28, 0]
nagents: 1

agents:
- name: "example_vessel"
  type: "usv"
  active_dof: [1, 1, 0, 0, 0, 1]
  U: 2.0
  maintain_speed: false
  geometry: "inputs/{name}/geometry.yml"
  inertia: "inputs/{name}/inertia.yml"
  hydrodynamics: "inputs/{name}/hydrodynamics.yml"
  control_surfaces: "inputs/{name}/control_surfaces.yml"
```



```
thrusters: "inputs/{name}/thrusters.yml"
initial_conditions: "inputs/{name}/initial_conditions.yml"
sensors: "inputs/{name}/sensors.yml"
guidance: "inputs/{name}/guidance.yml"
```

2. geometry.yml:

```
geometry_file: "inputs/example_vessel/geometry.gdf"
length: 5.0
breadth: 1.5
depth: 1.0

CO:
  position: [0.0, 0.0, 0.0]
  orientation: [0.0, 0.0, 0.0]

CG:
  position: [0.0, 0.0, 0.2]
  orientation: [0.0, 0.0, 0.0]

CB:
  position: [0.0, 0.0, -0.3]
  orientation: [0.0, 0.0, 0.0]

gyration: [1.5, 1.5, 0.8]
```

3. inertia.yml:

```
mass: 1000.0
buoyancy_mass: 1000.0
inertia_matrix: "None"
added_mass_matrix: "None"
```

4. hydrodynamics.yml:

```
dim_flag: false
hydra_file: "inputs/example_vessel/hydrodynamics.json"
coriolis_flag: true

X_u: -50.0
X_u_au: -100.0
Y_v: -200.0
```

```
Y_v_av: -400.0
Y_r: -50.0
N_v: -50.0
N_r: -300.0
N_r_ar: -400.0
```

5. **control_surfaces.yml:**

```
naca_file: "inputs/naca0015.csv"

control_surfaces:
- control_surface_id: 1
  control_surface_name: "Rudder"
  control_surface_location: [2.4, 0.0, 0.0]
  control_surface_orientation: [0, 0, 1.57]
  control_surface_area: 0.5
  control_surface_T: 0.5
  control_surface_delta_max: 35.0
  control_surface_deltad_max: 20.0
  control_surface_hydrodynamics: "None"
```

6. **thrusters.yml:**

```
thrusters:
- thruster_id: 1
  thruster_name: "Main Propeller"
  thruster_location: [-2.4, 0.0, 0.0]
  thruster_orientation: [0.0, 0.0, 0.0]
  D_prop: 0.4
  KT_at_J0: 0.5
  n_max: 1500
  tp: 0.05
```

7. **initial_conditions.yml:**

```
start_location: [0.0, 0.0, 0.0]
start_orientation: [0.0, 0.0, 0.0]
start_velocity: [0.0, 0.0, 0.0, 0.0, 0.0, 0.0]
use_quaternion: false
```

8. **sensors.yml:**

```
sensors:
  - sensor_type: "IMU"
    sensor_topic: "/vessel/imu"
    sensor_location: [0.0, 0.0, 0.1]
    sensor_orientation: [0.0, 0.0, 0.0]
    publish_rate: 100
    use_custom_covariance: true
    custom_covariance:
      orientation_covariance: [0.01, 0.0, 0.0, 0.0, 0.01, 0.0, 0.0, 0.0, 0.01]
      angular_velocity_covariance: [0.01, 0.0, 0.0, 0.0, 0.01, 0.0, 0.0, 0.0, 0.01]
      linear_acceleration_covariance: [0.05, 0.0, 0.0, 0.0, 0.05, 0.0, 0.0, 0.0, 0.05]
```

9. guidance.yml:

```
waypoints:
  - [0.0, 0.0, 0.0]
  - [50.0, 0.0, 0.0]
  - [50.0, 50.0, 0.0]
  - [0.0, 50.0, 0.0]
  - [0.0, 0.0, 0.0]
waypoints_type: XYZ
```

ROS2 Architecture

Overview

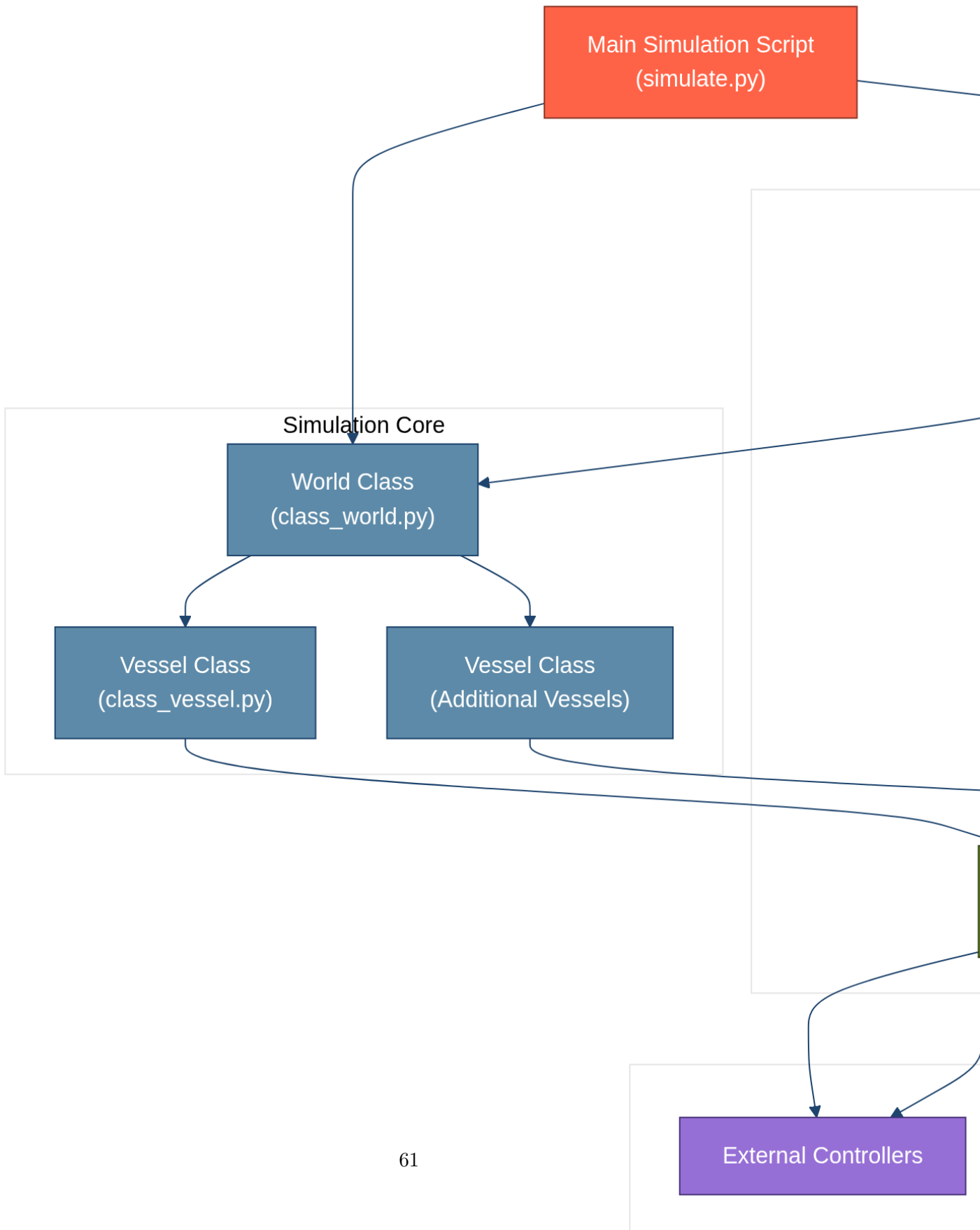
The Panisim runs on the Robot Operating System 2 (ROS2), enabling distributed simulation, external control, visualization, and data recording. The ROS2 integration architecture follows a modular design pattern that separates the simulation core from the communication layer. The Docker container of the simulator runs ROS2 Humble.

Tip

The ROS2 integration allows the Panisim to run completely isolated from controllers or guidance systems, thus enabling you to code your own controllers or guidance systems to interact with the simulator.

Architecture

The ROS2 integration consists of several key components that work together to bridge the simulation with the ROS2 ecosystem:



Explanation of the Architecture Flow:

- **Main Simulation Script (`simulate.py`):** Initiates the simulation by:
 - Calling `rclpy.init()` to initialize the ROS2 middleware
 - Creating a `World` instance by calling `World(config_file_path)`
 - Instantiating a `World_Node` that wraps the simulation in ROS2
 - Calling `world.start_vessel_ros_nodes(world_node)` to connect vessels to ROS2
- **World Class (`class_world.py`):**
 - Maintains multiple vessel instances using `vessels = []`
 - Processes world input with `process_world_input(world_file)`
 - Creates vessels via `Vessel(vessel_params, hydrodynamic_data, vessel_id)`
 - Updates simulation state with `step()` method that calls `vessel.step()` for each vessel
- **World_Node Class (`class_world_node_ros2.py`):**
 - Inherits from ROS2 Node class
 - Creates a timer with `create_timer(1/self.rate, self.world.step)` to drive simulation
- **Vessel_Pub_Sub Class (`class_vessel_pub_sub_ros2.py`):**
 - Creates publishers, subscribers, and timers for each vessel
 - Publishes vessel state and sensor data
 - Handles actuator commands via subscribers

Key Components

Main Simulation Script (`simulate.py`)

The entry point for running Panisim with ROS2 integration. This script:

1. Initializes the ROS2 environment
2. Creates the World simulation instance
3. Sets up the World_Node for ROS2 communication
4. Establishes connections between simulation and ROS2
5. Runs the ROS2 spin loop in a separate thread

```
def main():
    # Creates an object of class 'World'
    rclpy.init()
    world = World('/workspaces/mavlab/inputs/simulation_input.yml')
    world_node = World_Node(world_rate=1/world.dt)

    world.node = world_node
    world.start_vessel_ros_nodes(world_node)

    # Run ROS on a separate thread
    ros_thread_instance = threading.Thread(target=ros_thread, args=(world_node,))
    ros_thread_instance.start()

    # Prints available ROS2 topics and usage information
    # ...
```

The `ros_thread` function is defined as:

```
def ros_thread(node):
    rclpy.spin(node)
    rclpy.shutdown()
```

This function handles the ROS2 event loop, allowing callbacks to be processed in the background while the main thread can perform other tasks.

World Class (`class_world.py`)

The World class serves as the simulation environment container, managing:

- Multiple vessel instances within a unified simulation
- Initialization from configuration files
- Simulation time stepping across all vessels
- Global parameters like GPS datum and world boundaries

```
class World():
    terminate = False          # Flag indicating simulation termination
    vessels = []               # List of Vessel objects
    nvessels = 0               # Number of vessels
    size = np.zeros(3)         # Size of the world (X-Y-Z)
    gps_datum = None           # GPS reference point
    node = None                # ROS2 node reference
```

```

dt = 0.01                                # Simulation time step

def __init__(self, world_file=None):
    """Initialize the World object from configuration file"""
    if world_file is not None:
        self.process_world_input(world_file)

def step(self):
    """Advance the simulation by one time step"""
    for vessel in self.vessels:
        vessel.step()

```

The `process_world_input` method reads the simulation configuration from YAML files:

```

def process_world_input(self, world_file=None):
    try:
        sim_params, agents = read_input(world_file)
        self.size = np.array(sim_params['world_size'])
        self.nvessels = sim_params['nagents']
        self.gps_datum = np.array(sim_params['gps_datum'])
        agent_count = 0
        self.dt = sim_params['time_step']

        for agent in agents[0:self.nvessels]:
            vessel_config = agent['vessel_config']
            hydrodynamic_data = agent['hydrodynamics']
            self.vessels.append(Vessel(vessel_params=vessel_config,
                                       hydrodynamic_data=hydrodynamic_data,
                                       vessel_id=agent_count))

            agent_count += 1
    except yaml.YAMLError as exc:
        print(exc)
        exit()

```

The `World` class includes a method to start the ROS2 nodes for all vessels:

```

def start_vessel_ros_nodes(self, world_node):
    """Initialize ROS2 nodes for all vessels in the world"""
    for vessel in self.vessels:
        vessel.vessel_node = Vessel_Pub_Sub(vessel, world_node)

```


World_Node Class (class_world_node_ros2.py)

The World_Node class is a ROS2 node wrapper for the World class:

```
class World_Node(Node):
    rate = None
    def __init__(self, world_rate=100, world_file=None):
        super().__init__('world')
        self.rate = world_rate
        self.world = World(world_file)
        self.create_timer(1/self.rate, callback=self.world.step)
```

This class: - Inherits from the ROS2 Node class - Creates a timer that triggers the simulation step function at a specified rate - Provides the ROS2 context for the World class

The `create_timer` method sets up a timer that calls the `World.step()` method at regular intervals determined by `world_rate`. This is what drives the simulation forward in time while maintaining synchronization with the ROS2 ecosystem.

Vessel_Pub_Sub Class (class_vessel_pub_sub_ros2.py)

The Vessel_Pub_Sub class implements the ROS2 communication interface for vessel objects:

```
class Vessel_Pub_Sub():
    def __init__(self, vessel, world_node):
        """Initialize the vessel interface"""
        self.world_node = world_node
        self.vessel = vessel
        self.vessel_id = vessel.vessel_id
        self.vessel_name = vessel.vessel_name
        self.topic_prefix = f'{self.vessel_name}_{self.vessel_id:02d}'

        # Initialize publishers, subscribers, and timers
        # ...
```

This class:

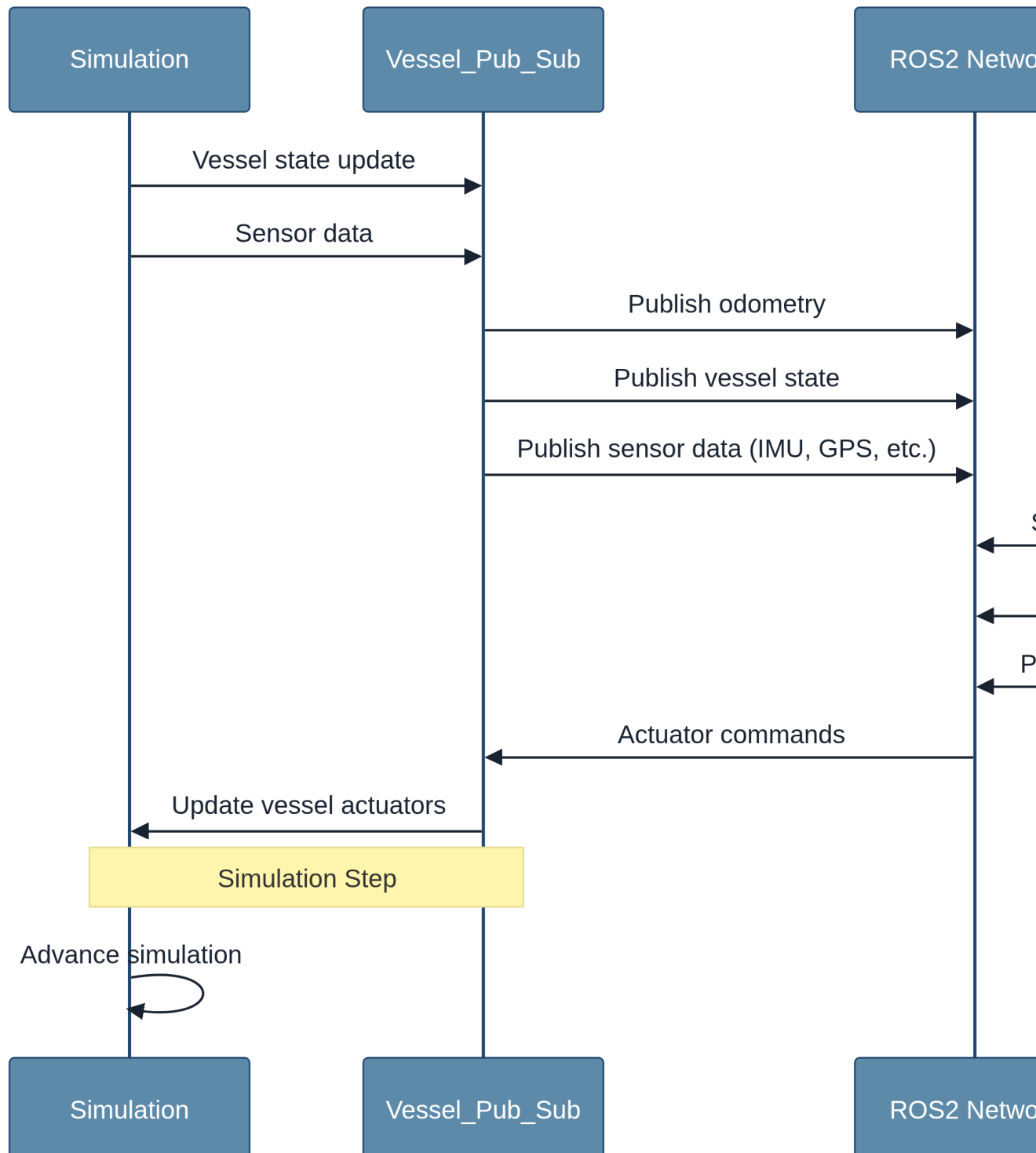
- Creates and manages ROS2 publishers for vessel state and sensor data
- Sets up subscribers for actuator commands
- Handles message conversion between simulator data and ROS2 messages
- Manages sensor data publication at appropriate rates

Key initialization methods include:

- Setting up control surface and thruster mappings
- Creating sensor objects and their publishers
- Initializing odometry and vessel state publishers
- Creating actuator command subscribers

Message Flow

The communication between simulation components and the ROS2 ecosystem follows this message flow:



Detailed Function Call Sequence:

1. Vessel State Update:

- `vessel.step()` in `World.step()` updates the vessel state
- This updates `vessel.current_state`, which is accessed by publishers

2. Sensor Data Generation:

- Sensor objects call `get_measurement()` to generate simulated sensor readings based on vessel state

3. Publishing Data to ROS2:

- `publish_odometry()` is called by a timer to publish vessel position and orientation
- `publish_vessel_state()` is called by a timer to publish complete state vector
- `_publish_sensor()` is called by sensor-specific timers to publish sensor data

4. External Nodes Subscribing:

- External ROS2 nodes subscribe to topics using `ros2 topic echo` or programmatically

5. Control Commands:

- External nodes publish to `/<vessel_name>_<id>/actuator_cmd`
- `actuator_callback(msg)` processes these commands

6. Updating Actuator States:

- `vessel.delta_c` (control surfaces) and `vessel.n_c` (thrusters) are updated
- These affect the vessel dynamics in the next simulation step

7. Simulation Step:

- The timer in `World_Node` triggers `world.step()`
- This advances all vessels' states using their dynamics models

Topics and Messages

Each vessel in the simulation publishes data on several ROS2 topics:

Standard Topics

Topic	Message Type	Description
<code>/<vessel_name>_<id>/odometry</code>	<code>nav_msgs/Odometry</code>	Vessel position, orientation, and velocities

Topic	Message Type	Description
/<vessel_name>_<id>/vessel_state	std_msgs/Float64MultiArray	Complete vessel state including actuators
/<vessel_name>_<id>/interfaces/Actuator	interfaces/Actuator	Commands for control surfaces and thrusters

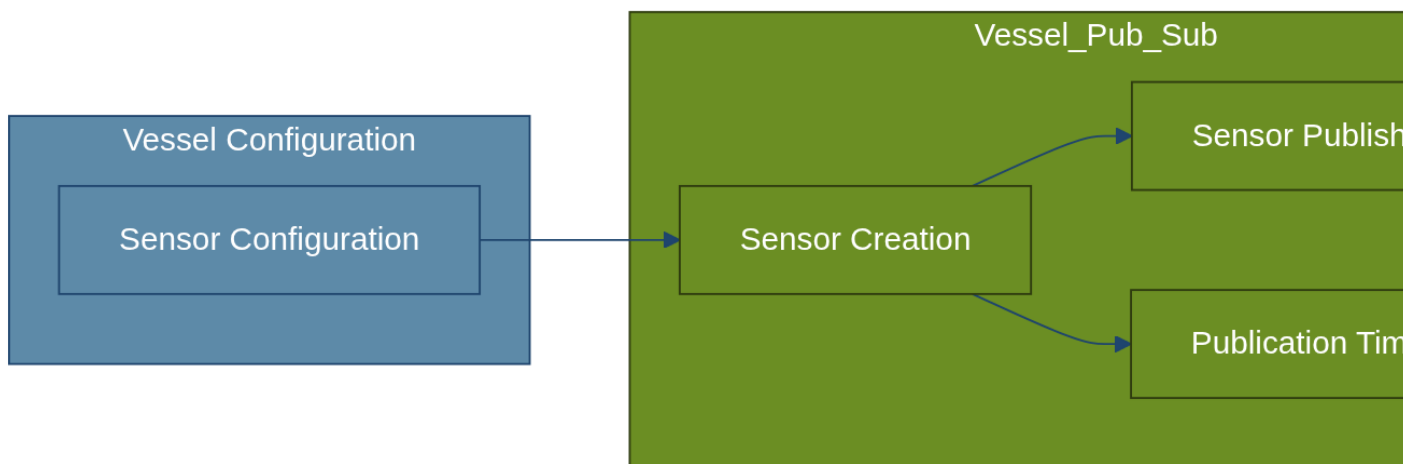
Sensor Topics

Each sensor configured for a vessel publishes on its own topic:

Sensor	Topic	Message Type
IMU	/<vessel_name>_<id>/sensor_msgs/Imu	sensor_msgs/Imu
GPS	/<vessel_name>_<id>/sensor_msgs/NavSatFix	sensor_msgs/NavSatFix
DVL	/<vessel_name>_<id>/interfaces/DVL	interfaces/DVL
UWB	/<vessel_name>_<id>/geometry_msgs/PoseWithCovarianceStamped	geometry_msgs/PoseWithCovarianceStamped

Sensor Integration

Sensors are created and managed through the Vessel_Pub_Sub class:



Detailed Function Calls in Sensor Integration:

1. SensorConfig to SensorCreation:

- During `Vessel_Pub_Sub` initialization, it reads sensor configurations from `vessel.vessel_config['sensors']`
- For each sensor configuration, it calls `create_sensor(sensor_config, vessel_id, topic_prefix, self)`

2. SensorCreation to Publishers and Timers:

- After creating a sensor object, `Vessel_Pub_Sub` creates a publisher using:

```
self.world_node.create_publisher(self._get_msg_type(sensor.sensor_type), sensor_topi
```

- It also creates a timer for each sensor:

```
self.world_node.create_timer(1/sensor.rate, lambda s=sensor: self._publish_sensor(s
```

3. Timer to PublishSensor:

- Each timer periodically calls `_publish_sensor(sensor)` at the sensor's configured rate

4. PublishSensor to GetMeasurement:

- The `_publish_sensor` method calls `sensor.get_measurement()` to obtain the latest sensor reading
- Each sensor type has its own implementation of `get_measurement()`

5. PublishSensor to CreateMessage:

- After getting the measurement, `_publish_sensor` calls `_create_sensor_message(sensor_type, measurement)`
- This converts the measurement to the appropriate ROS2 message type

6. CreateMessage to Topics:

- Finally, the message is published to the ROS2 network:

```
s['pub'].publish(msg)
```

The process for sensor integration:

1. Sensor configurations are defined in the vessel parameters (input files `sensors.yml`)
2. The `Vessel_Pub_Sub` class creates sensor objects using the `create_sensor` factory function
3. For each sensor, a publisher and timer are created
4. The timer periodically calls the sensor's `get_measurement()` method
5. Measurements are converted to ROS2 messages and published

Key methods in the sensor integration:

```
def _publish_sensor(self, sensor):
    """Publish sensor data."""
    measurement = sensor.get_measurement()
    msg = self._create_sensor_message(sensor.sensor_type, measurement)

    # Find the publisher for this sensor
    for s in self.sensors:
        if s['sensor'] == sensor:
            s['pub'].publish(msg)
            break

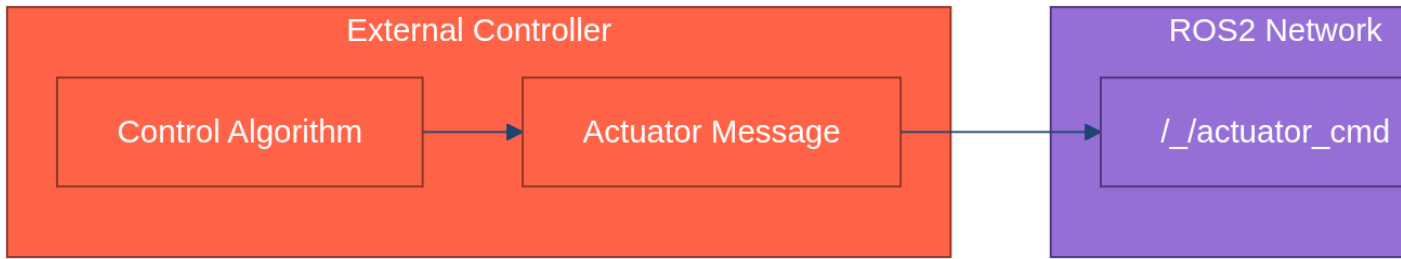
def _create_sensor_message(self, sensor_type, measurement):
    """Create a ROS message for sensor data."""
    current_time = self.world_node.get_clock().now().to_msg()

    # Create appropriate message based on sensor type
    if sensor_type == 'IMU':
        msg = Imu()
        # Fill message fields
    elif sensor_type == 'GPS':
        msg = NavSatFix()
        # Fill message fields
    # ... other sensor types

    return msg
```

Actuator Control

Vessels can be controlled through ROS2 by sending actuator commands:



Detailed Function Calls in Actuator Control:

1. ControlAlgorithm to ActuatorMsg:

- External control algorithms compute desired actuator values
- They create an `Actuator` message with the desired values

2. ActuatorMsg to ActuatorTopic:

- The controller publishes the message to the actuator command topic
- This is done using ROS2's `publisher.publish(msg)` mechanism

3. ActuatorTopic to Subscriber:

- The subscriber in `Vessel_Pub_Sub` receives the message
- This was set up during initialization with:

```
self.actuator_sub = self.world_node.create_subscription(  
    Actuator,  
    f'{self.topic_prefix}/actuator_cmd',  
    self.actuator_callback,  
    1  
)
```

4. Subscriber to Callback:

- The subscriber triggers `actuator_callback(msg)` in `Vessel_Pub_Sub`

5. Callback to ControlSurfaces and Thrusters:

- The callback parses the message and updates vessel control values:


```
# For control surfaces
self.vessel.delta_c[idx] = value * np.pi / 180.0

# For thrusters
self.vessel.n_c[idx] = value
```

Actuator Message Structure (Actuator.msg)

The `Actuator.msg` is a custom message type used for controlling vessel actuators. Its structure is:

```
std_msgs/Header header      # Standard header with timestamp
string[] actuator_names     # Array of actuator identifiers
float64[] actuator_values   # Array of corresponding values
```

The actuator names use a prefixing convention: - **cs**_N for control surfaces (e.g., **cs**_1 for the first control surface) - **th**_N for thrusters (e.g., **th**_1 for the first thruster)

Values for control surfaces are specified in **degrees** (converted to radians internally) and values for thrusters are in **RPM** (Revolutions Per Minute).

Example usage in `class_vessel_pub_sub_ros2.py`:

```
def actuator_callback(self, msg):
    """Handle actuator command messages."""
    if len(msg.actuator_names) != len(msg.actuator_values):
        self.world_node.get_logger().warn('Mismatch between actuator IDs and values length')
        return

    for actuator_id, value in zip(msg.actuator_names, msg.actuator_values):
        try:
            if actuator_id.startswith('cs_'):
                # Handle control surface (convert degrees to radians)
                if actuator_id in self.control_surface_ids:
                    idx = self.control_surface_ids[actuator_id]
                    self.vessel.delta_c[idx] = value * np.pi / 180.0
                else:
                    self.world_node.get_logger().warn(f'Unknown control surface ID: {actuator_id}')
            elif actuator_id.startswith('th_'):
                # Handle thruster (RPM value)
```

```

        if actuator_id in self.thruster_ids:
            idx = self.thruster_ids[actuator_id]
            self.vessel.n_c[idx] = value
        else:
            self.world_node.get_logger().warn(f'Unknown thruster ID: {actuator_id}')

    else:
        self.world_node.get_logger().warn(f'Invalid actuator ID format: {actuator_id}')
except (ValueError, IndexError):
    self.world_node.get_logger().warn(f'Unknown actuator ID: {actuator_id}')

```

During initialization, `Vessel_Pub_Sub` builds mappings between actuator IDs (as defined in `control_surfaces.yml` and `thrusters.yml`) and their indices:

```

# Store actuator IDs with type prefixes
self.control_surface_ids = {} # Maps 'cs_id' to index
self.thruster_ids = {}       # Maps 'th_id' to index

for idx, cs in enumerate(self.control_surfaces):
    cs_id = cs.get('control_surface_id', idx+1)
    self.control_surface_ids[f'cs_{cs_id}'] = idx

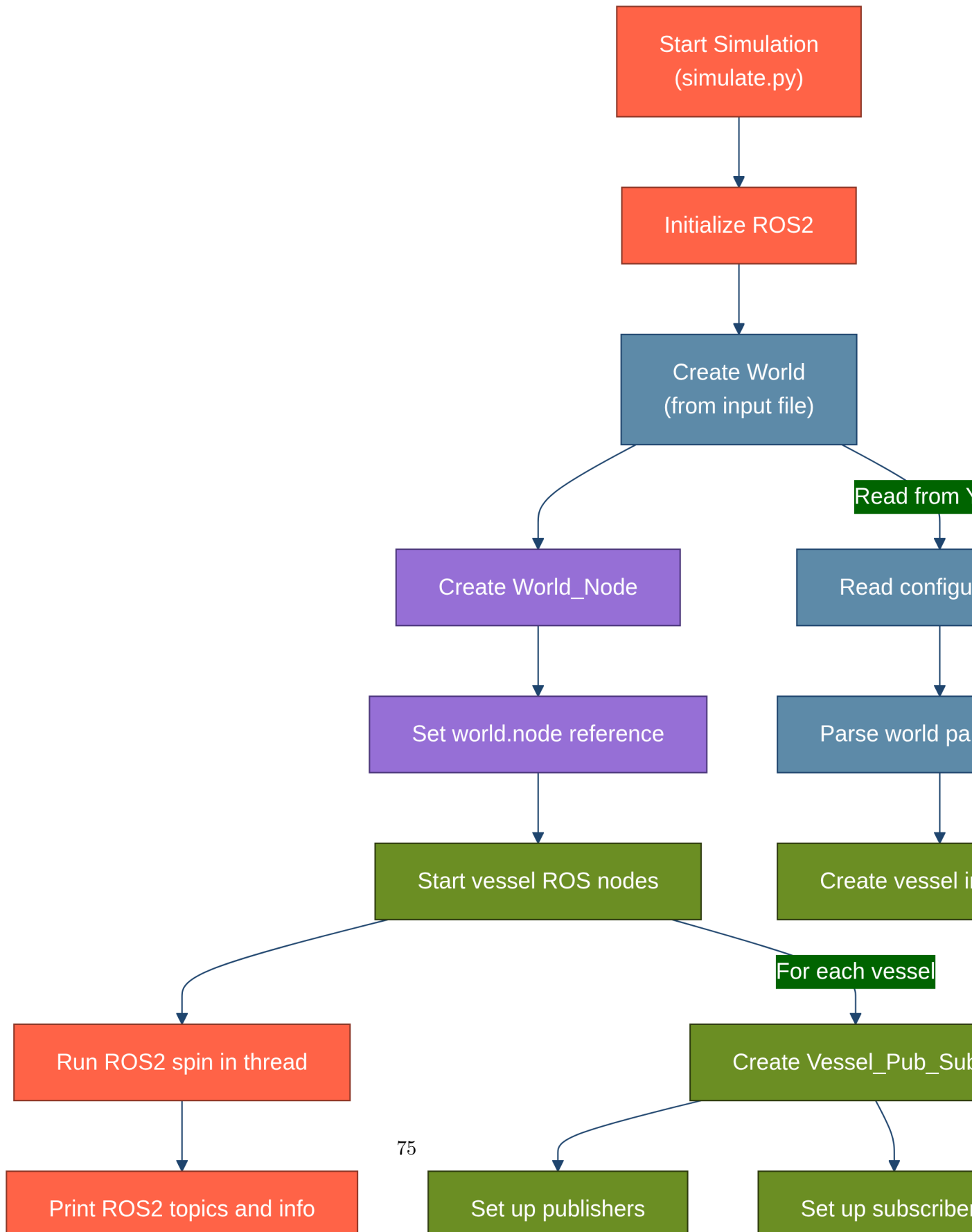
for idx, th in enumerate(self.thrusters):
    th_id = th.get('thruster_id', idx+1)
    self.thruster_ids[f'th_{th_id}'] = idx

```

These mappings ensure that actuator commands are properly routed to the correct control surfaces and thrusters in the vessel's internal state.

World and Vessel Population

The process of populating the world with vessels follows these steps:



Detailed Function Calls in World and Vessel Population:

1. Start to Init:

- `simulate.py` starts and calls `rclpy.init()` to initialize the ROS2 middleware
- This initializes the ROS2 context for the application

2. Init to CreateWorld:

- `World('/workspaces/mavlab/inputs/simulation_input.yml')` is called
- This creates the World instance and loads configuration

3. CreateWorld to ReadConfig:

- `World.__init__` calls `self.process_world_input(world_file)`
- `process_world_input` calls `read_input(world_file)` to parse YAML

4. ReadConfig to ParseWorld:

- `process_world_input` extracts world parameters like:
 - `self.size = np.array(sim_params['world_size'])`
 - `self.nvessels = sim_params['nagents']`
 - `self.gps_datum = np.array(sim_params['gps_datum'])`

5. ParseWorld to CreateVessels:

- `process_world_input` creates vessel instances:

```
self.vessels.append(Vessel(vessel_params=vessel_config,
                           hydrodynamic_data=hydrodynamic_data,
                           vessel_id=agent_count))
```

6. CreateWorld to CreateNode:

- `World_Node(world_rate=1/world.dt)` is called
- This creates a ROS2 node wrapper for the World

7. SetNode to StartVessels:

- `world.start_vessel_ros_nodes(world_node)` is called
- This iterates through vessels and creates ROS2 interfaces

8. StartVessels to CreateVesselPubSub:

- `Vessel_Pub_Sub(vessel, world_node)` is called for each vessel
- Each vessel gets its own ROS2 communication interface

9. CreateVesselPubSub to SetupPublishers:

- Publishers are created for each vessel's odometry and state:

```

self.odometry = {
    'pub': self.world_node.create_publisher(Odometry, f'{self.topic_prefix}/odometry', 10),
    'timer': self.world_node.create_timer(self.vessel.vessel_config['time_step'], self.odometry_callback)
}

```

10. CreateVesselPubSub to SetupSubscribers:

- Subscribers are created for actuator commands:

```

self.actuator_sub = self.world_node.create_subscription(
    Actuator,
    f'{self.topic_prefix}/actuator_cmd',
    self.actuator_callback,
    1
)

```

11. CreateVesselPubSub to SetupSensors:

- Sensors are created and configured based on vessel configuration:

```

sensor = create_sensor(sensor_config, self.vessel_id, self.topic_prefix, self)
self.sensors.append({
    'sensor': sensor,
    'pub': self.world_node.create_publisher(self._get_msg_type(sensor.sensor_type), f'{self.topic_prefix}/{sensor.name}_pub', 10),
    'timer': self.world_node.create_timer(1/sensor.rate, lambda s=sensor: self._publish_sensor_data(s))
})

```

12. StartVessels to SpinThread:

- `threading.Thread(target=ros_thread, args=(world_node,))` is created and started
- This runs `rospy.spin(node)` in a separate thread

13. SpinThread to PrintInfo:

- Available ROS2 topics are listed and helpful usage information is printed

Key steps in this process:

1. The simulation begins in `simulate.py`
2. ROS2 is initialized
3. A World object is created from the input configuration file
4. A World_Node is created and linked to the World object
5. Vessel ROS nodes are started for each vessel in the world
6. For each vessel, a Vessel_Pub_Sub object is created, which:
 - Sets up publishers for odometry and vessel state

- Sets up subscribers for actuator commands
- Creates sensor objects based on the vessel configuration
- Sets up timers for publishing sensor data

Running the Simulation

To run the MAV simulator with ROS2 integration:

```
# Navigate to the workspace
cd /path/to/makara

# Build ROS2 packages
colcon build

# Source the workspace
source install/setup.bash

# Run the simulator
ros2 run mav_simulator simulate
```

After starting the simulation, you can interact with it using standard ROS2 commands:

```
# List all available topics
ros2 topic list

# Subscribe to vessel odometry
ros2 topic echo /<vessel_name>_<id>/odometry_sim

# Publish control commands
ros2 topic pub /<vessel_name>_<id>/actuator_cmd interfaces/Actuator \
"{header: {stamp: {sec: 0}}, actuator_names: ['cs_1', 'th_1'], actuator_values: [15.0, 1200.0]}
```

Best Practices

- **Launch Files:** Create launch files for different simulation scenarios. There are some example launch files in the `/workspace/mavlab/ros2_ws/src/mav_simulator/mav_simulator/launch` folder.

Vessel Configurator

Overview

The Vessel Configurator is a light weight web-based application that provides an intuitive interface for configuring, and generating input files for the Panisim simulator. This tool eliminates the need to manually create YAML configuration files, offering instead a visual approach to visualize a vessel in 3D and configure its components.

Previously, without the web GUI, one would have to manually enter entries into the YAML files, which was very time consuming and error prone. The major parameters entered in the YAML files are the position and orientation of the various components such as thrusters, control surfaces, sensors etc. With the Web GUI, you simply load a FBX file into the configurator, select your components and the software logs the position and orientation of the components automatically still allowing you to make modifications. You can select the Centre of Buoyancy, Centre of Gravity and Body Centre visually and Panisim takes care of the frame of reference conversion (to the body frame) automatically.

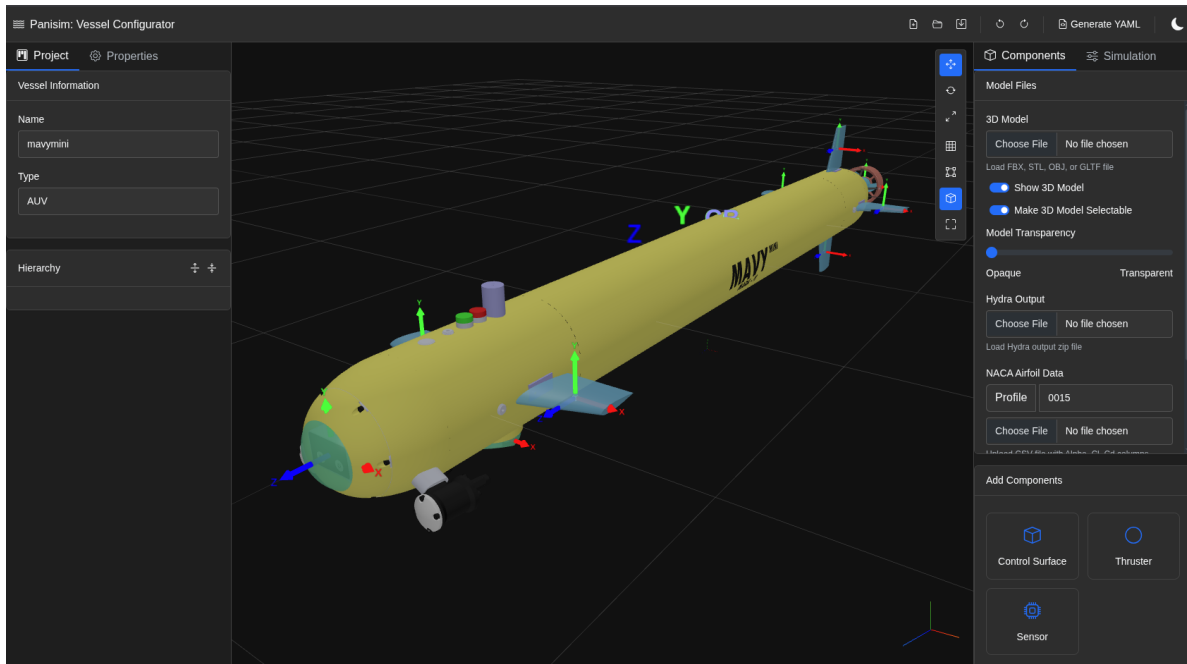


Figure 5: An example configuration in the vessel configurator

Tip

The vessel configurator was largely built using AI!

Note

The configurator is still needs development and some features may be unstable. The `guidance.yml` and `control.yml` files will still need to be manually edited.

Tutorial

To Quickstart and playaround, you can load the pre-configured `mavymini_config.json` vessel model in the `/inputs/example configurations` folder. Or follow the step below to configure your own vessel.

Step 1: Load a Vessel Model

1. Open the vessel configurator in your web browser

2. Click on the “Choose Vessel” button
3. Select the FBX file of the vessel model you want to configure

Now you should see the vessel model in the 3D view.

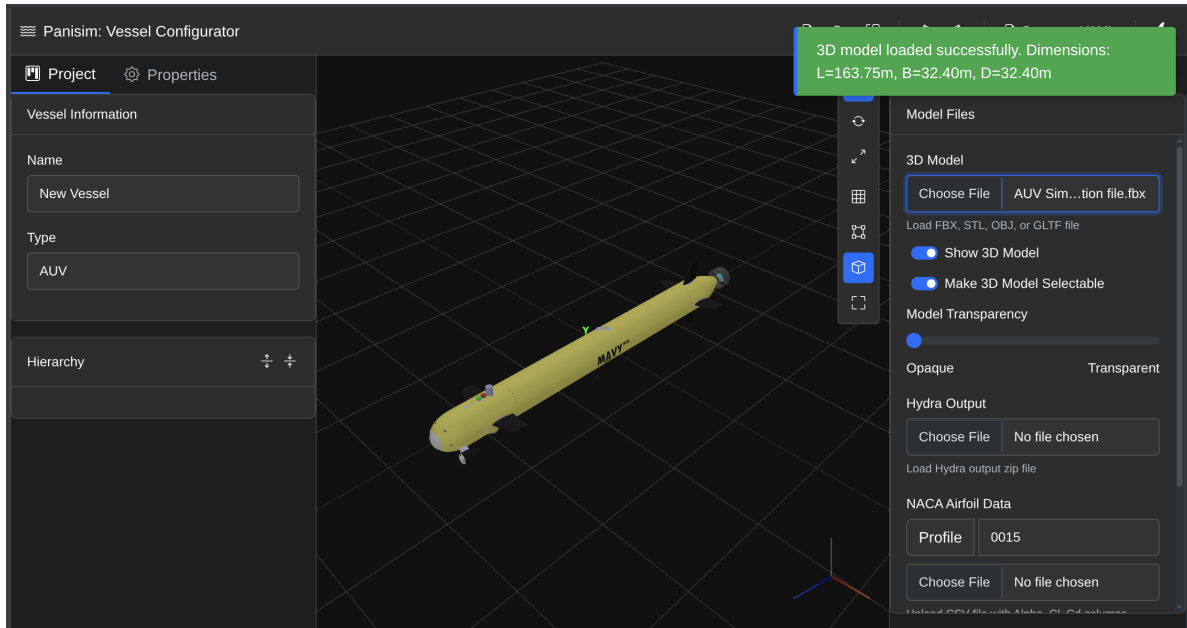


Figure 6: Loaded Vessel

Tip

The software automatically calculates the length, breadth and depth of the vessel from the FBX file via a bounding box. if possible, you should have your vessel in the configuration used in marine vessels, i.e X axis along the length of the vessel pointing forward, Y axis along the breadth pointing to the starboard and Z axis along the depth pointing downwards.

Step 2: Configure the Vessel

1. Select the component (Control Surfaces, Thrusters, or Sensors) you want to configure. It will be slightly highlighted on selection.

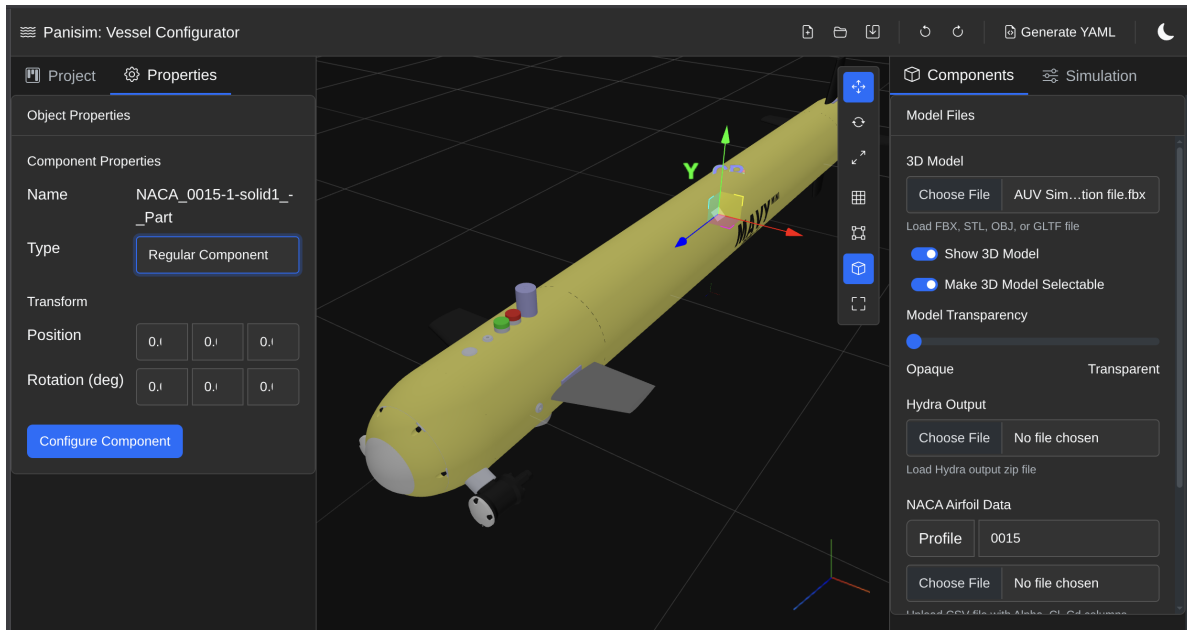


Figure 7: Selected Component

2. From the left sidebar, switch to the Properties tab and select the component type.
3. Configuration modal would appear on selecting the component type.

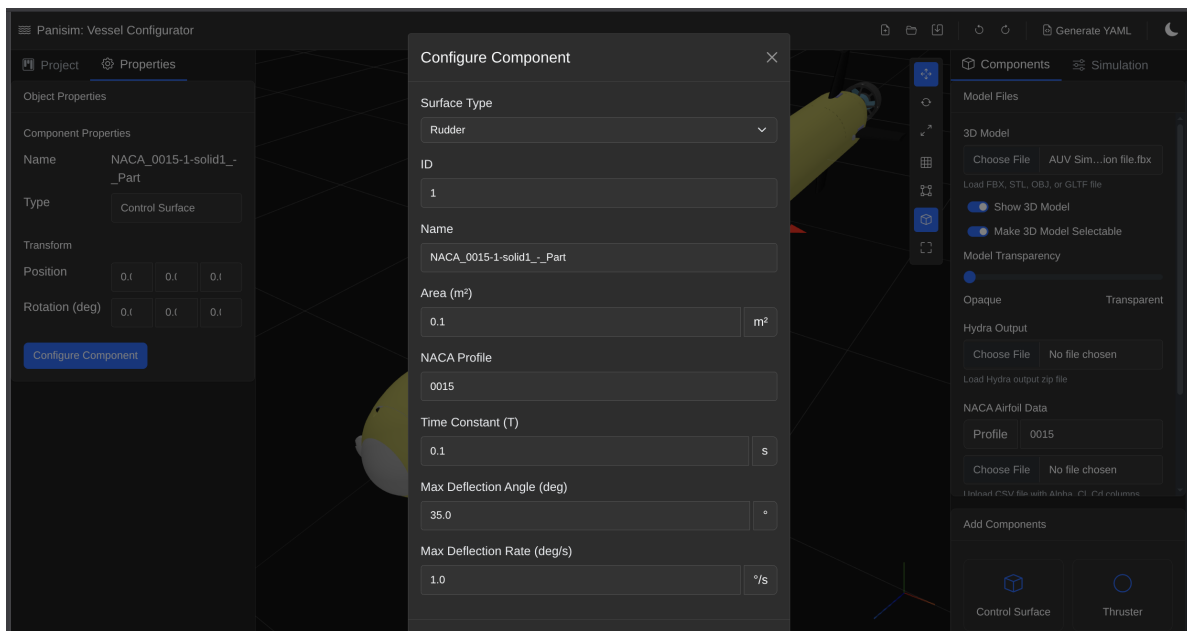


Figure 8: Component Configuration Modal

4. Enter the required parameters and click on the “Apply” button.
5. The component will then be configured. This can be confirmed if you see a coordinate system assigned to your vessel.

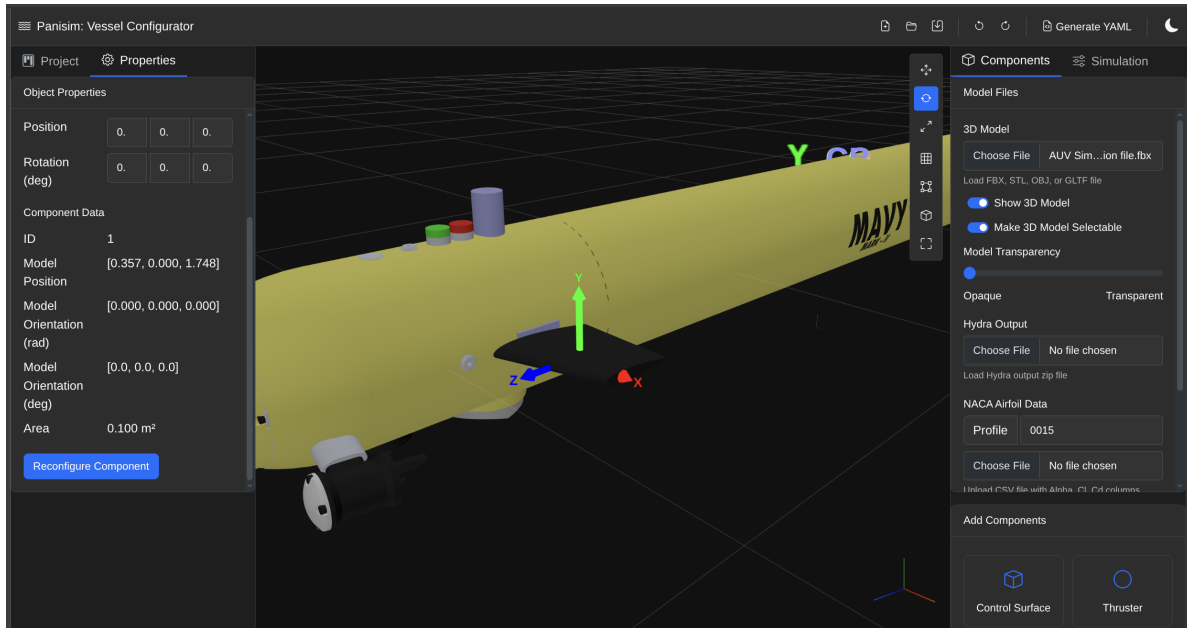


Figure 9: Component Configured

6. You should see the position and orientation in the properties tab.
7. You can select the coordinate and transform it in the 3D view based on where you want to place it.

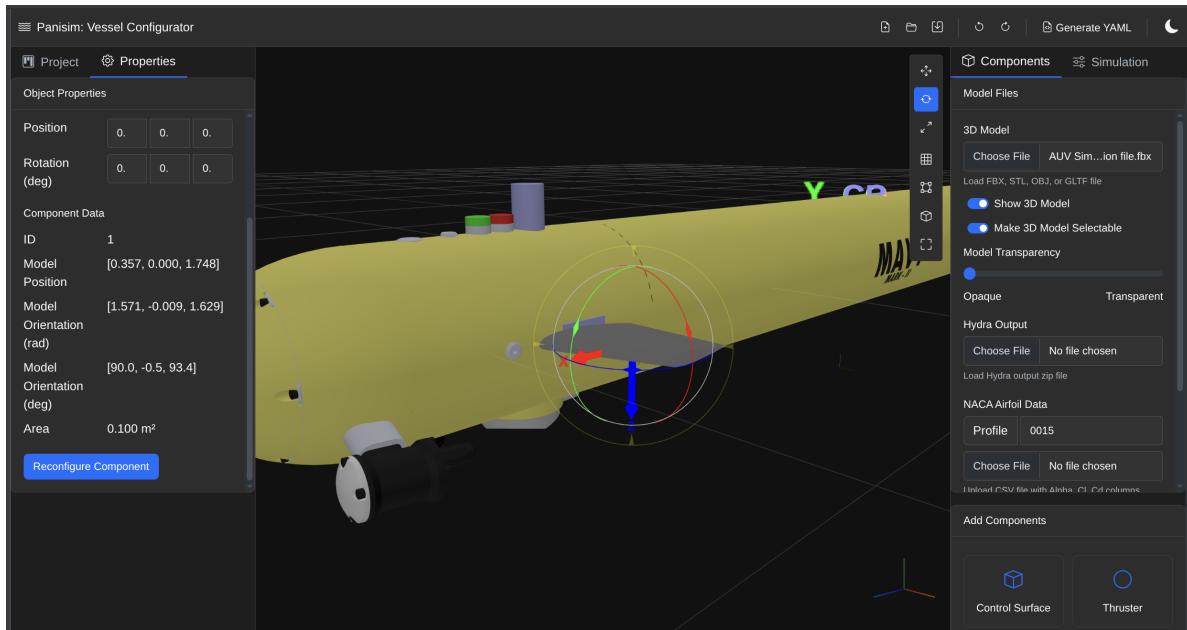


Figure 10: Component Configured

For sensors, the orientation have to be how you have mounted it on the vessel.

For control surfaces, the X axis needs to be along the chord, Y axis pointing along the span inwards to the vessel and Z axis completing the right hand coordinate system.

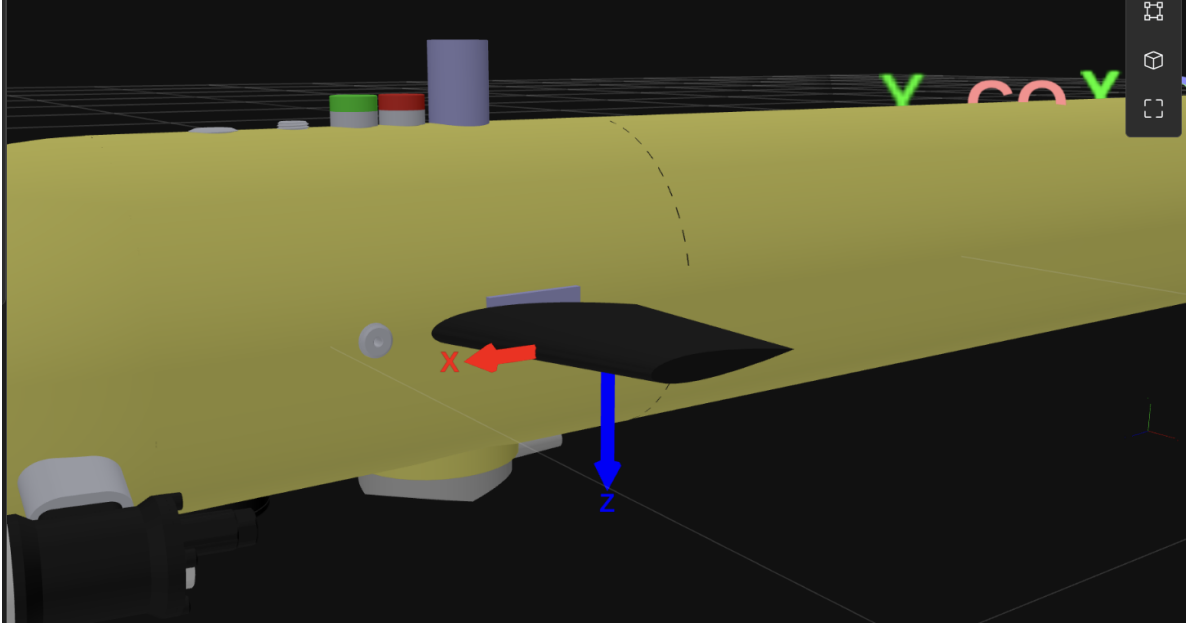


Figure 11: Control Surface

For thrusters, the X axis needs to be along the axis of the thruster, Y axis pointing towards the starboard and Z axis completing the right hand coordinate system.

Everything follows the NED (North, East, Down) coordinate system.

Step 3: Placing the vessel center points

1. Make the vessel transparent to see the inside. And untoggle the “Make 3D Model Selectable” option. This will allow you to select the centre points and transform them.

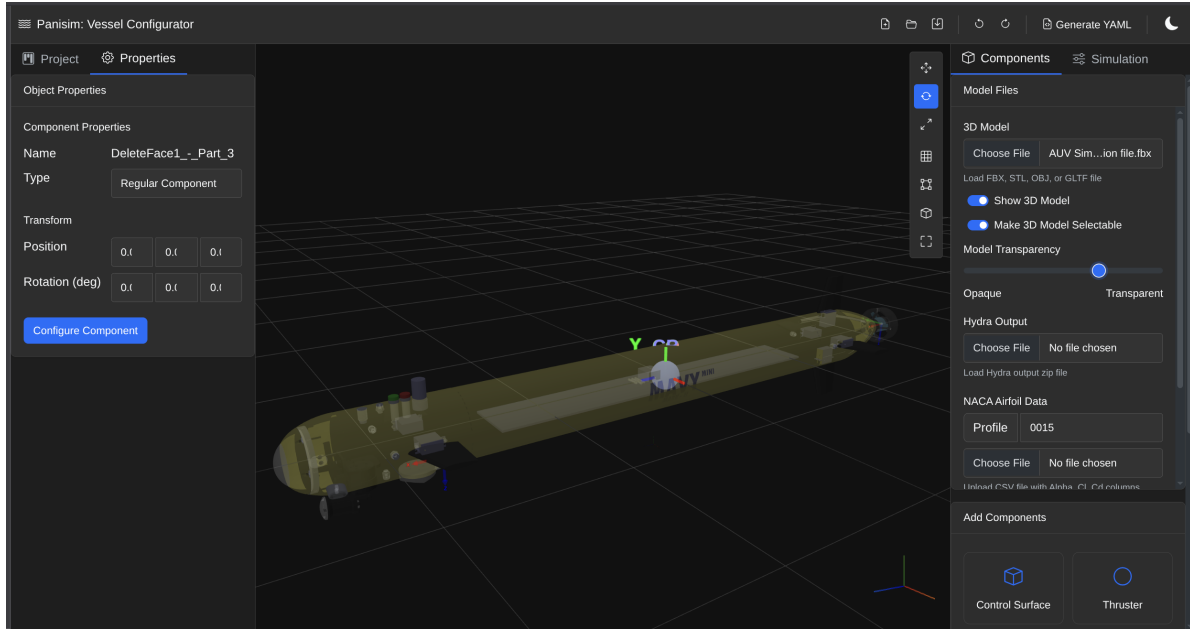


Figure 12: Transparent Vessel

2. You can select each of the center points and transform them to the desired location.
3. You can also manually enter the coordinates in the **Edit Geometry Parameters** in the right sidebar components tab.
4. The Body Centre [CO], is the point with respect to which all other coordinates are defined. And therefore it must be placed in the NED configuration.

Step 4: Edit Hydrodynamic Coefficients

1. Edit the hydrodynamic coefficients in the **Edit Hydrodynamic Coefficients** in the right sidebar components tab. Follow the instructions in the modal to enter the hydrodynamic coefficients.

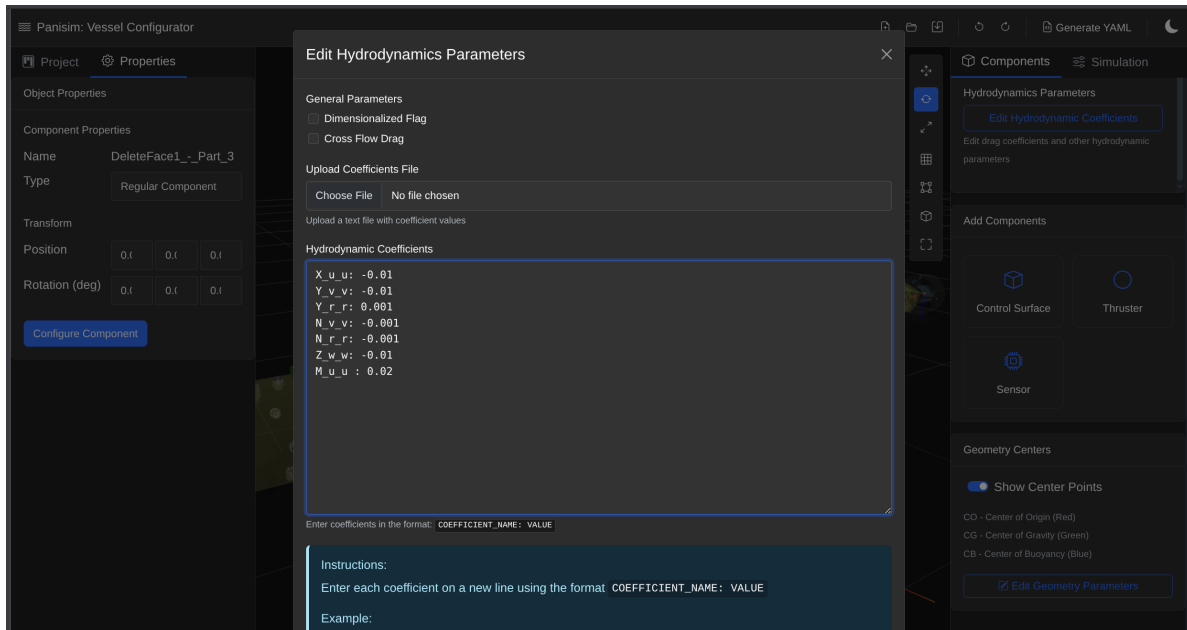


Figure 13: Hydrodynamic Coefficients

Step 5: Simulation settings

1. Edit all the parameters in the **Simulation** tab in the right sidebar.

i Note

In the **Advanced Simulation Parameters**, you can visually select the geofence boundaries. This also can be further extended to select waypoints in a future release.

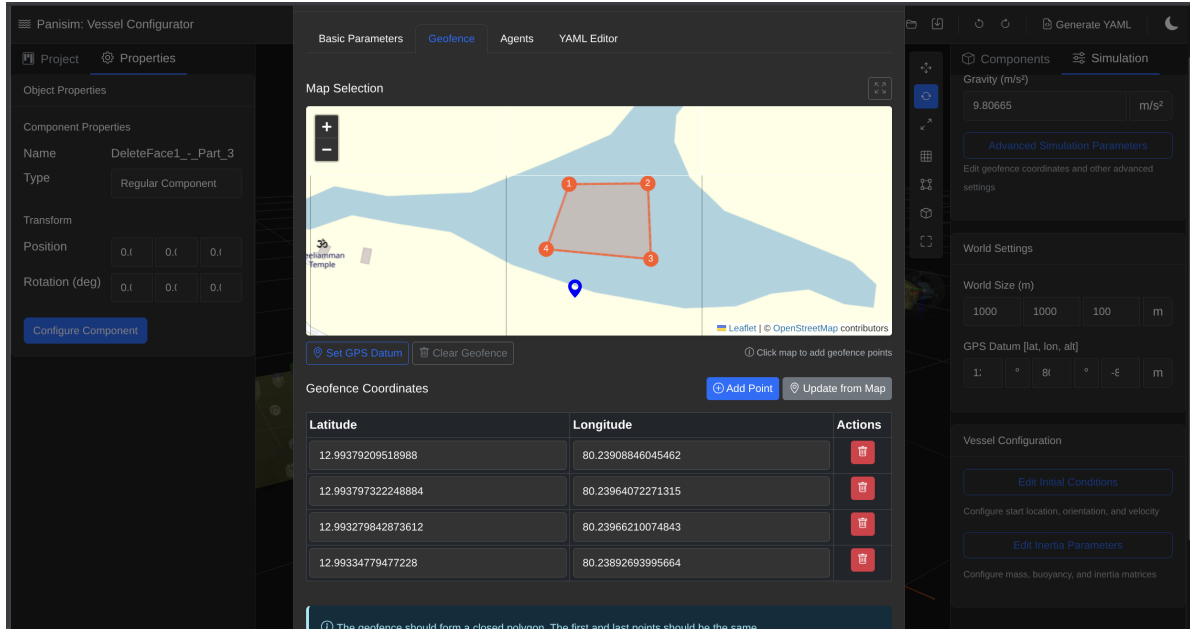


Figure 14: Geofence

Step 6: Upload the Hydra output file (optional, if you are not using cross-flow drag)

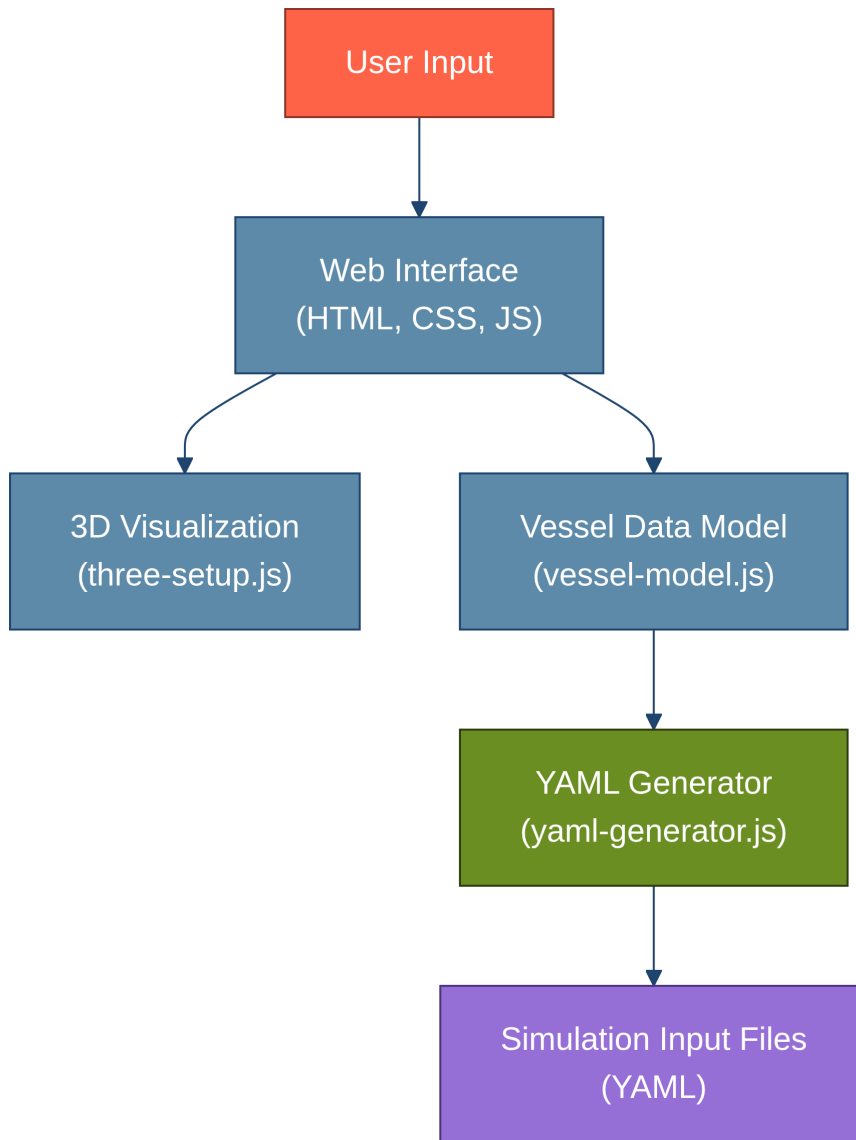
You can get your Hydra output file from the HyDRA website. And upload the zip folder in the web gui. The generated input file will then contain your Hydra file with correct path.

Step 7: Generate the YAML files

1. Click on the “Generate YAML” button in the toolbar.
2. The YAML files will be generated and downloaded as a zip file.
3. Unzip the file and place it in the **inputs** directory.

And there you have it! You have successfully configured your vessel and generated the required input files for the Panisim simulator. Some configuration file may require manual edits. Feel free to play around with the Web GUI and extend its functionality.

Software Architecture



The Vessel Configurator provides a complete workflow for creating and configuring marine vessels with the following key features:

- **Parameter Configuration:** Intuitive interfaces for setting physical properties and simulation parameters, GPS waypoints.
- **Component Management:** Add, edit, and position thrusters, control surfaces, and sensors

- **YAML Generation:** Automatic generation of properly formatted configuration files for the simulator
- **Validation:** Built-in checks to ensure valid component ID configuration

Core Components

1. User Interface (`index.html`, `styles.css`, `themes.css`)

- Implements a responsive, CAD-style interface with Bootstrap 5.3.2 framework integration
- Features a component hierarchy with:
 - Primary toolbar containing application controls (new/load/save/undo/redo)
 - Split-pane workspace with resizable panels using Split.js
 - Left sidebar with tabbed project explorer and properties inspector
 - 3D viewport with transform controls and visualization options
 - Right sidebar with component addition and configuration panels
 - Modal dialogs for advanced configuration options and YAML preview
- Supports context-sensitive property editing with dynamic form generation
- Implements theme switching functionality with CSS variables for light/dark modes
- Maintains responsive layout through Bootstrap grid system and custom flex containers

2. 3D Visualization Engine (`three-setup.js`)

- Implements a Three.js-based rendering system with WebGL acceleration
- Configures high-performance renderer with antialiasing, physically correct lighting, and shadow mapping
- Manages scene graph with hierarchical component structure and parent-child relationships
- Features multiple coordinate systems (world, vessel-local, component-local)
- Implements interactive controls with:
 - OrbitControls for camera navigation (pan, rotate, zoom)
 - TransformControls for direct manipulation (translate, rotate, scale)
 - Raycaster-based object selection with visual feedback
- Provides real-time synchronization between 3D objects and data model properties
- Implements custom visual feedback systems including:
 - Color-coded component highlighting for selection state
 - Local axes visualization for component orientation
 - Text sprite labeling for identification and measurements
 - Dynamic material updates for selection and hover states
- Optimizes rendering performance using:
 - Request animation frame with conditional rendering

- Efficient mesh creation with geometry instancing
- Adaptive resolution scaling based on device capabilities
- Conditional shadow casting based on object importance

3. Vessel Data Model (`vessel-model.js`)

- Implements a comprehensive object-oriented data structure with 1156+ lines of structured code
- Maintains vessel configuration as a deeply nested JavaScript object with strong typing conventions
- Organizes data in specialized subsystems:
 - Physical properties (dimensions, mass, centers)
 - Hydrodynamic coefficients (added mass, damping, restoring forces)
 - Component collections (control surfaces, thrusters, sensors)
 - Simulation parameters (time step, environment, boundary conditions)
- Provides transaction-based modification methods with validation:
 - Addition, update, and removal operations for all component types
 - Auto-incrementing ID allocation for component tracking
 - Reference integrity management between components
 - Unit conversion and normalization for consistent data representation
- Implements bidirectional mapping between model data and 3D objects using UUID tracking
- Maintains persistence through JSON serialization/deserialization with state version control
- Supports incremental updates through partial property modification methods

4. YAML Generator (`yaml-generator.js`)

- Implements specialized transformation engine for converting vessel model to simulation-compatible YAML
- Features robust numerical formatting with:
 - Configurable precision control (4-6 decimal places based on parameter type)
 - Unit annotation through comments
 - Scientific notation for small coefficient values
 - Array formatting with dimension-appropriate representation
- Organizes output into standard simulation file hierarchy:
 - `geometry.yml` with vessel dimensions and center points
 - `hydrodynamics.yml` with coefficient grouping by motion direction (X/Y/Z/K/M/N)
 - `inertia.yml` with mass matrix and added mass coefficients
 - `control_surfaces.yml` with NACA profile properties and dynamics
 - `thrusters.yml` with propulsion characteristics and placement
 - `initial_conditions.yml` with startup position and velocity
 - `sensors.yml` with instrumentation configuration and noise models

- Generates Docker-compatible file paths for simulator integration (`/workspaces/mavlab/inputs/...`)
- Performs semantic validation with appropriate warning generation
- Uses JSZip library for creating multi-file archives with proper directory structure

5. Utility Modules

- `gdf-loader.js`: Implements parser for Geometric Data Files (GDF) with:
 - Triangulated mesh conversion
 - Hydrodynamic coefficient extraction
 - Center of buoyancy calculation from mesh geometry
- `controls.js`: Extends Three.js transform controls with:
 - Custom snapping behavior for precise positioning
 - Event handling for synchronized model updates
 - Specialized axis constraints for different component types
- External library integration:
 - Three.js (r128) for 3D visualization
 - JSZip (3.10.1) for configuration packaging
 - js-yaml (4.1.0) for YAML parsing/generation
 - Bootstrap (5.3.2) for UI components
 - Leaflet (1.9.4) for geofencing and waypoint mapping
 - FileSaver (2.0.5) for client-side file downloads

Simulation Visualization Dashboard

Overview

Panisim provides a web-based visualization dashboard to monitor vessel simulation in real-time. The dashboard offers a comprehensive interface for monitoring vessel state, visualizing trajectories, viewing sensor data, and interacting with the simulation.

The visualization system consists of two main components:

1. **2D Dashboard** - A data-rich interface showing vessel state, charts, and control inputs
2. **3D Visualization** - A three-dimensional view of the vessel in its environment with position and orientation

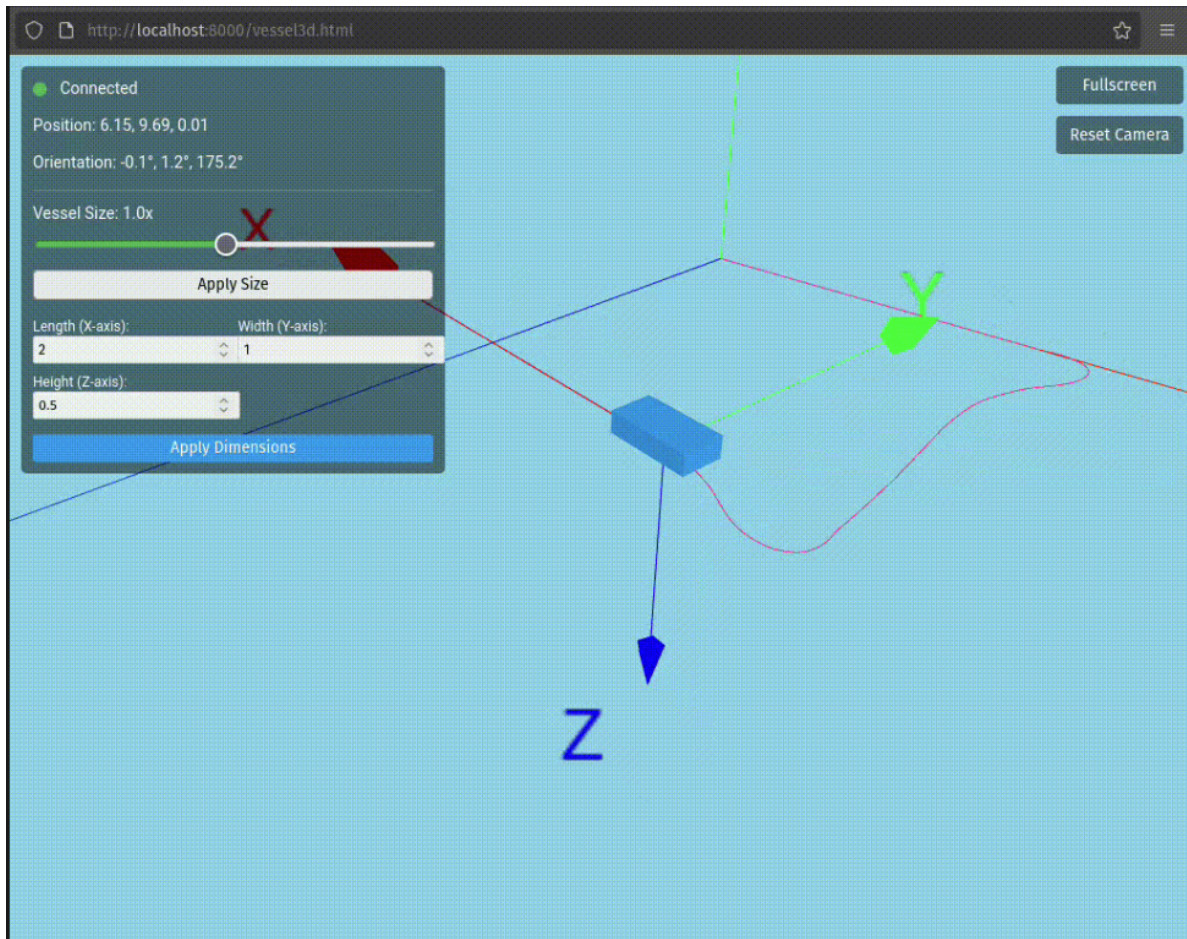
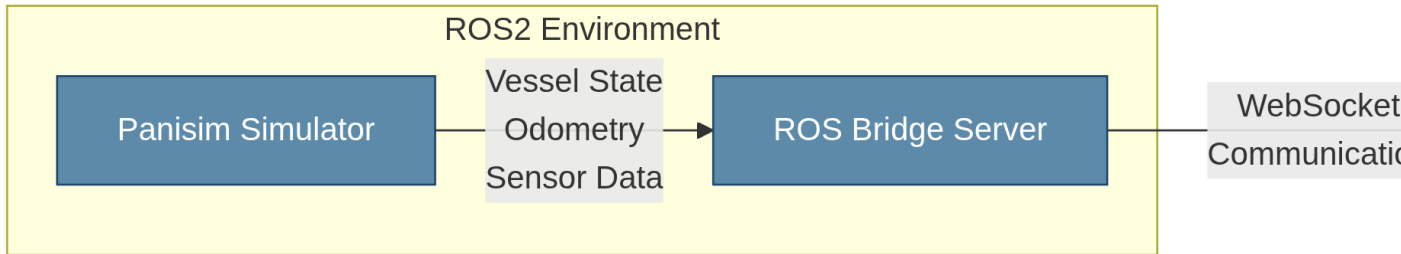


Figure 15: Way-point trackingSimulation Visualization



Starting the Visualization

The web interface is automatically started when launching the Panisim simulator. It uses:

1. A lightweight HTTP server to host the web files
2. The ROS Bridge server for WebSocket communication with ROS2

The visualization launches when running the simulation:

```
ros2 launch gnc gnc_launch.py
```

Once started, you can access the dashboard by opening your web browser:

- **Main Dashboard:** <http://localhost:8000/>
- **3D Visualization:** <http://localhost:8000/vessel3d.html>

Main Dashboard

The main dashboard (index.html) provides a comprehensive overview of the vessel's state, sensor data, and control inputs.

Key Features

1. **Vessel Selection:** Switch between different vessels in the simulation
2. **Connection Status:** Real-time display of connectivity to the ROS2 environment
3. **Vessel Information:** Display of key vessel parameters (position, velocity, orientation)
4. **Data Visualizations:** Real-time charts for:
 - Vessel path and trajectory
 - Velocities (linear and angular)
 - Orientation (roll, pitch, yaw)
 - Control surface positions
 - Thruster states

Dashboard Layout

The dashboard is organized in a responsive grid layout with cards for different data categories:

Section	Description
Vessel Info	Basic vessel parameters including position, velocities, and orientation
Path Visualization	2D trajectory visualization showing vessel movement in X-Y plane
Vessel Velocities	Charts for surge (u), sway (v), and heave (w) velocities
Vessel Angular Rates	Charts for roll rate (p), pitch rate (q), and yaw rate (r)
Orientation	Charts for roll (), pitch (), and yaw () angles
Control Surfaces	Visual representation of control surface positions
Thrusters	Visual representation of thruster RPMs

Real-time Data Updates

The dashboard updates in real-time with configurable update rates. Data collection begins automatically when connecting to a vessel and includes:

- Subscribe to `/[vessel_name]/odometry_sim` for position and velocity data
- Subscribe to `/[vessel_name]/vessel_state` for complete vessel state including actuators
- Optional subscription to additional sensor topics

3D Visualization

The 3D visualization (vessel3d.html) provides an interactive three-dimensional view of the vessel in its environment.

Key Features

1. **Real-time Movement:** The vessel model moves and rotates according to simulation data
2. **Trajectory Path:** A colored trail showing the vessel's path through the environment
3. **Customizable View:** Camera controls for rotating, panning, and zooming
4. **Vessel Dimensions:** Adjustable vessel size and proportions to match different vessel types
5. **Full-screen Mode:** Expand the visualization to utilize the entire screen

Controls

The 3D view offers several interactive controls:

- **Orbit:** Click and drag to rotate the camera around the vessel
- **Pan:** Right-click and drag (or middle-click and drag) to move the camera
- **Zoom:** Scroll wheel to zoom in and out
- **Reset Camera:** Button to restore the default camera position
- **Fullscreen:** Button to toggle fullscreen mode

Vessel Model Customization

You can adjust the vessel's appearance through the control panel:

- **Vessel Size:** Overall scaling of the vessel model
- **Dimensions:** Individual control over length, width, and height proportions
- **Apply Changes:** Button to update the model with the new dimensions

Integration with ROS2

The visualization dashboard communicates with ROS2 using the ROS Bridge Server, which provides WebSocket connectivity between the ROS2 environment and web clients.

Communication Flow

1. **ROS2 Topics to WebSockets:** The ROS Bridge Server converts ROS2 topic messages to WebSocket messages
2. **Web Client Processing:** JavaScript code processes the WebSocket messages and updates the visualizations
3. **Bidirectional Communication:** The system supports both subscribing to and publishing ROS2 topics

Required ROS2 Components

To enable the visualization, the launch file includes:

```
# HTTP server for web files
ExecuteProcess(
  cmd=['python3', '-m', 'http.server', '8000', '--directory', web_build_dir],
  name='http_server',
  output='screen'
),

# ROS Bridge server for WebSocket communication
IncludeLaunchDescription(
  AnyLaunchDescriptionSource(rosbridge_launch)
),
```

Topic Subscriptions

The visualization automatically discovers vessel topics by querying available ROS2 topics. It looks for the following patterns:

- `/<vessel_name>/odometry_sim`: For vessel position and velocity
- `/<vessel_name>/vessel_state`: For complete vessel state including actuators

Implementation Details

The visualization dashboard is built using modern web technologies:

- **Chart.js**: For responsive, interactive data visualization
- **three.js**: For 3D rendering of the vessel
- **roslibjs**: For ROS2 communication via WebSockets
- **Responsive Design**: Automatically adapts to different screen sizes

Key JavaScript Components

The system is organized into the following components:

1. **VesselManager**: Handles vessel discovery, selection, and configuration
2. **VesselVisualizer**: Manages data collection, processing, and visualization
3. **3D Rendering**: Handles the three-dimensional visualization of vessels

Vessel State Representation

The dashboard visualizes the complete vessel state model:

- **Position**: x, y, z (in meters)
- **Linear Velocity**: u, v, w (in m/s)
- **Angular Velocity**: p, q, r (in rad/s)
- **Orientation**: ϕ, θ, ψ (in degrees)
- **Actuators**: Control surface angles (in degrees) and thruster RPMs

Customization Options

The visualization can be customized in several ways:

1. **Update Rates**: Chart refresh rates can be adjusted
2. **Data Window**: The amount of historical data displayed in charts
3. **Vessel Configuration**: The number of control surfaces and thrusters displayed
4. **3D Model**: Vessel dimensions and appearance
5. **Chart Scaling**: Auto-rescaling and manual zoom/pan on charts

Troubleshooting

Connection Issues

If the dashboard shows “Disconnected” status:

1. Ensure the ROS Bridge server is running
2. Check that WebSocket port 9090 is accessible
3. Verify the simulator is publishing required topics

Missing Vessel Data

If vessel data isn't displaying:

1. Select a vessel from the dropdown menu
2. Check ROS2 topics are being published:

```
ros2 topic list | grep vessel
```

3. Verify message publication with:

```
ros2 topic echo /<vessel_name>/odometry_sim
```

Browser Compatibility

The visualization works best with modern browsers:

- Google Chrome
- Mozilla Firefox
- Microsoft Edge
- Brave Browser

Conclusion

The Panisim visualization dashboard provides a comprehensive interface for monitoring and interacting with simulated vessels. With its real-time updates, interactive charts, and 3D visualization, it offers a powerful tool for understanding vessel behavior and validating custom control or guidance algorithms.

Example - Creating a ROS2 Waypoint Tracking Controller

Overview

This example demonstrates how to create a custom ROS2 package to interact with the Panisim simulator. We'll build a Guidance, Navigation, and Control (GNC) package that implements a waypoint tracking PID controller for marine vessels. This example covers:

1. Creating a ROS2 package structure
2. Implementing core controller logic
3. Setting up ROS2 nodes to interface with the simulator
4. Building and launching the complete system

Tip

All the below codes and folders are already implemented in the Repository. This example is only for the reference and to show how you can configure your own ROS2 package to interact with the Panisim simulator.

Creating the ROS2 Package

1. Package Structure

First, let's create a new ROS2 package called `gnc` in the `ros2_ws/src` directory:

```
cd ros2_ws/src
ros2 pkg create --build-type ament_python gnc --dependencies rclpy nav_msgs geometry_msgs in
```

This creates a basic package structure:

```
gnc/  
  gnc/  
    __init__.py  
    package.xml  
    setup.py  
    setup.cfg
```

2. Update Package Dependencies

Edit `package.xml` to include all required dependencies:

```
<?xml version="1.0"?>  
<?xml-model href="http://download.ros.org/schema/package_format3.xsd" schematypens="http://www.ros.org/xml-schemas/package_format3.xsd"?>  
<package format="3">  
  <name>gnc</name>  
  <version>0.0.1</version>  
  <description>Guidance, Navigation and Control package for Panisim</description>  
  <maintainer email="example@example.com">user</maintainer>  
  <license>Apache License 2.0</license>  
  
  <depend>rclpy</depend>  
  <depend>nav_msgs</depend>  
  <depend>geometry_msgs</depend>  
  <depend>std_msgs</depend>  
  <depend>interfaces</depend>  
  <depend>mav_simulator</depend>  
  
  <exec_depend>ros2launch</exec_depend>  
  
  <export>  
    <build_type>ament_python</build_type>  
  </export>  
</package>
```

3. Configure Setup Scripts

Update `setup.py` to include our entry points and specify dependencies:

```
from setuptools import setup  
import os  
from glob import glob
```

```

package_name = 'gnc'

setup(
    name=package_name,
    version='0.0.1',
    packages=[package_name],
    data_files=[
        ('share/ament_index/resource_index/packages',
         ['resource/' + package_name]),
        ('share/' + package_name, ['package.xml']),
        (os.path.join('share', package_name, 'launch'), glob('launch/*.py')),
        (os.path.join('share', package_name, 'web'), glob('web/**/*', recursive=True)),
    ],
    install_requires=['setuptools'],
    zip_safe=True,
    maintainer='user',
    maintainer_email='example@example.com',
    description='Guidance, Navigation and Control package for Panisim',
    license='Apache License 2.0',
    tests_require=['pytest'],
    entry_points={
        'console_scripts': [
            'gc = gnc.guidance_control:main',
            'nav = gnc.navigation:main',
        ],
    },
)

```

4. Create Directory Structure

Set up the complete directory structure:

```

mkdir -p gnc/launch
mkdir -p gnc/web

```

Implementing the Controller Logic

1. Control Module

First, create `module_control.py` in the `gnc` package to implement the PID control logic:

```
# ros2_ws/src/gnc/gnc/module_control.py
import numpy as np

"""
Rudder control module for vessel simulation.

This module provides rudder control strategies for vessel steering with PID controllers.
"""

def ssa(ang, deg=False):
    """
    Smallest signed angle that lies between -pi and pi.
    If deg is True, the angle is assumed to be in degrees and is converted to radians.
    If deg is False, the angle is assumed to be in radians.
    """
    if deg:
        ang = (ang + 180) % (360.0) - 180.0
    else:
        ang = (ang + np.pi) % (2 * np.pi) - np.pi
    return ang

def clip(val, min_val, max_val):
    """
    Clip a value to a range.
    """
    return max(min_val, min(val, max_val))

def pid_control(t, state, waypoints, waypoint_idx, ye_int=0.0):
    """
    Implement a PID control strategy to follow the waypoints.

    Args:
        t (float): Current simulation time [s]
        state (ndarray): Current vessel state vector
        waypoints (ndarray): Waypoints array
        waypoint_idx (int): Current waypoint index
    """
```



```

    ye_int (float): Integral term of cross-track error

Returns:
    float: Commanded rudder angle in radians
    float: Cross-track error
    int: Next waypoint index
"""

# Check if we have reached the last waypoint
if waypoint_idx == len(waypoints):
    return 0.0, 0.0, waypoint_idx

# Current state
u, v, r = state[3], state[4], state[5]
x, y, psi = state[6], state[7], state[11]

# Current and previous waypoints
wp_xn, wp_yn, _ = waypoints[waypoint_idx]
wp_xn1, wp_yn1, _ = waypoints[waypoint_idx - 1]

# Calculate path line equation: ax + by + c = 0
a = -(wp_yn1 - wp_yn)
b = (wp_xn1 - wp_xn)
c = -wp_yn1*b-a*wp_xn1

# Path direction unit vector
wp_unit_vec = np.array([wp_xn - wp_xn1, wp_yn - wp_yn1, 0.0])
wp_unit_vec = wp_unit_vec / np.linalg.norm(wp_unit_vec)

# Calculate cross-track error
ye = -(a*x + b*y + c)/np.sqrt(a**2 + b**2)

# Path direction angle
pi_p = np.angle(wp_unit_vec[0] + 1j * wp_unit_vec[1])

# Outer loop PID gains
Kpo = 0.6 # Proportional gain
Kio = 0.05 # Integral gain

# Desired heading (outer loop control)
psid = pi_p + Kpo*(-ye) + Kio*(-ye_int)

```

```

# Inner loop PID gains
Kpi = 0.7 # Proportional gain
Kdi = 0.5 # Derivative gain

# Rudder command (inner loop control)
delta_c = Kpi*ssa(psid - psi) + Kdi*(0 - r)

# Clip the rudder angle
delta_c = clip(delta_c, -35*np.pi/180, 35*np.pi/180)

# Distance to waypoint
wp_dist = np.linalg.norm(np.array([x - wp_xn, y - wp_yn, 0.0]))
if wp_dist < 0.5: # 0.5m threshold
    waypoint_idx += 1

# Return negative delta_c (due to sign convention of rudder angle)
return -delta_c, ye, waypoint_idx

```

2. Guidance Control Node Class

Next, create `class_guidance_control.py` to implement the ROS2 node class:

```

# ros2_ws/src/gnc/gnc/class_guidance_control.py
from rclpy.node import Node
from nav_msgs.msg import Odometry
from interfaces.msg import Actuator
from geometry_msgs.msg import PoseArray, Pose
from mav_simulator.module_kinematics import quat_to_eul
import gnc.module_control as con
import numpy as np

class GuidanceControl(Node):
    def __init__(self, vessel):
        super().__init__('guidance_controller')

        self.vessel = vessel

        # Configure topics based on vessel name and ID
        self.odom_topic = f'{self.vessel.vessel_name}_{self.vessel.vessel_id:02d}/odometry_s'
        self.actuator_topic = f'{self.vessel.vessel_name}_{self.vessel.vessel_id:02d}/actuator'
        self.waypoints_topic = f'{self.vessel.vessel_name}_{self.vessel.vessel_id:02d}/waypoints'

```

```

# Create subscriber for odometry
self.odom_sub = self.create_subscription(
    Odometry,
    self.odom_topic,
    self.odom_callback,
    10)

# Create publisher for rudder command
self.actuator_pub = self.create_publisher(
    Actuator,
    self.actuator_topic,
    10)

# Create publisher for waypoints
self.waypoints_pub = self.create_publisher(
    PoseArray,
    self.waypoints_topic,
    10)

# Set up timer to publish waypoints
self.timer = self.create_timer(1.0, self.publish_waypoints)

# Read waypoints from vessel configuration
self.waypoints = np.array(self.vessel.vessel_config['guidance']['waypoints'])
self.waypoints_type = self.vessel.vessel_config['guidance']['waypoints_type']
self.waypoint_idx = 1

self.waypoints_data = {
    'waypoints': self.waypoints,
    'waypoints_type': self.waypoints_type
}

# Initialize time
self.start_time = self.get_clock().now()

# Initialize integrator
self.ye_int = 0.0
self.te_int = 0.0

# Select control mode
self.control_mode = con.pid_control

```

```

self.get_logger().info('Guidance Controller initialized')

def publish_waypoints(self):
    """Publish waypoints for visualization"""
    if not self.waypoints_data:
        return

    pose_array = PoseArray()
    pose_array.header.stamp = self.get_clock().now().to_msg()
    pose_array.header.frame_id = "map"

    for waypoint in self.waypoints_data.get('waypoints', []):
        pose = Pose()
        pose.position.x = float(waypoint[0])
        pose.position.y = float(waypoint[1])
        pose.position.z = float(waypoint[2])
        pose_array.poses.append(pose)

    self.waypoints_pub.publish(pose_array)

def odom_callback(self, msg):
    """Process odometry data and calculate control commands"""
    # Get current time in seconds
    current_time = self.get_clock().now()
    t = (current_time - self.start_time).nanoseconds / 1e9

    # Extract orientation quaternion
    quat = np.array([
        msg.pose.pose.orientation.w,
        msg.pose.pose.orientation.x,
        msg.pose.pose.orientation.y,
        msg.pose.pose.orientation.z
    ])

    # Convert quaternion to Euler angles
    eul = quat_to_eul(quat, order='ZYX')

    # Create state vector from odometry
    state = np.array([
        msg.twist.twist.linear.x,
        msg.twist.twist.linear.y,
        msg.twist.twist.linear.z,

```

```

        msg.twist.twist.angular.x,
        msg.twist.twist.angular.y,
        msg.twist.twist.angular.z,
        msg.pose.pose.position.x,
        msg.pose.pose.position.y,
        msg.pose.pose.position.z,
        eul[0],
        eul[1],
        eul[2],
        0.0 # rudder angle
    ])

    # Calculate control command
    rudder_cmd, ye, wp_idx = self.control_mode(t, state, self.waypoints, self.waypoint_idx)

    # Handle waypoint transitions
    if wp_idx != self.waypoint_idx:
        self.get_logger().info(f'Waypoint {self.waypoint_idx} reached')
        self.waypoint_idx = wp_idx
        self.ye_int = 0.0 # Reset integral term

    # Loop back to first waypoint when all are visited
    if self.waypoint_idx >= len(self.waypoints):
        self.get_logger().info('***All waypoints reached***')
        self.waypoint_idx = 1

    # Calculate time difference for integration
    dt = t - self.te_int
    if dt > 0: # Avoid division by zero or negative time
        # Update the integral term
        self.ye_int += ye * dt
        self.te_int = t

    # Publish actuator command
    cmd_msg = Actuator()
    cmd_msg.actuator_values = [float(rudder_cmd * 180 / np.pi)] # Convert to degrees
    cmd_msg.actuator_names = ['cs_1'] # Control surface 1
    cmd_msg.covariance = [0.0]
    self.actuator_pub.publish(cmd_msg)

```

3. Main Entry Point

Create `guidance_control.py` to serve as the entry point for the ROS2 node:

```
# ros2_ws/src/gnc/gnc/guidance_control.py
import rclpy
from gnc.class_guidance_control import GuidanceControl
from mav_simulator.class_world import World
from rclpy.executors import MultiThreadedExecutor

def main():
    # Initialize ROS2
    rclpy.init()

    # Load world and vessels from configuration
    world = World('/workspaces/mavlab/inputs/simulation_input.yml')
    vessels = world.vessels

    # Create multi-threaded executor
    executor = MultiThreadedExecutor()

    # Initialize guidance controllers for all vessels
    gcs = []
    for vessel in vessels:
        gcs.append(GuidanceControl(vessel))
        executor.add_node(gcs[-1])

    try:
        # Start spinning the nodes
        executor.spin()

    finally:
        # Clean up on shutdown
        executor.shutdown()

        for gc in gcs:
            gc.destroy_node()

        rclpy.shutdown()

if __name__ == '__main__':
    main()
```

4. Create a Navigation Node Placeholder

Add a simple placeholder for the navigation node to satisfy the launch file requirements:

```
# ros2_ws/src/gnc/gnc/navigation.py
import rclpy
from rclpy.node import Node

class Navigation(Node):
    def __init__(self):
        super().__init__('navigation')
        self.get_logger().info('Navigation node started')

def main():
    rclpy.init()
    node = Navigation()
    rclpy.spin(node)
    rclpy.shutdown()

if __name__ == '__main__':
    main()
```

Creating the Launch File

Create a launch file to start all components of the system:

```
# ros2_ws/src/gnc/launch/gnc_launch.py
import os
from launch import LaunchDescription
from launch_ros.actions import Node
from launch import LaunchDescription
from launch.actions import ExecuteProcess, IncludeLaunchDescription
from launch.launch_description_sources import AnyLaunchDescriptionSource
from launch_ros.substitutions import FindPackageShare
from launch.substitutions import PathJoinSubstitution
from launch_ros.actions import Node
from mav_simulator.class_world import World
from launch.substitutions import LaunchConfiguration
from launch.actions import GroupAction
from launch.launch_description_sources import PythonLaunchDescriptionSource
from launch_ros.actions import PushRosNamespace
```

```

def generate_launch_description():

    rosbridge_launch = PathJoinSubstitution([
        FindPackageShare('rosbridge_server'),
        'launch',
        'rosbridge_websocket_launch.xml'
    ])

    # Path to the web directory in gnc package
    web_dir = PathJoinSubstitution([
        FindPackageShare('gnc'),
        'web'
    ])

    web_build_dir = '/workspaces/mavlab/ros2_ws/src/mav_simulator/web'

    return LaunchDescription([
        Node(
            package='mav_simulator', # Replace with your package name
            executable='simulate', # Replace with your script name
            name='mavsim',
            output='screen'
        ),
        Node(
            package='gnc',
            executable='nav',
            name='nav',
            output='screen'
        ),
        Node(
            package='gnc',
            executable='gc',
            name='gc',
            output='screen'
        ),
        # Start the HTTP server to serve the web interface
        ExecuteProcess(
            cmd=['python3', '-m', 'http.server', '8000', '--directory', web_build_dir],
            name='http_server',
            output='screen'
        ),
        # Start the rosbridge server
    ])

```



```
    IncludeLaunchDescription(  
        AnyLaunchDescriptionSource(rosbridge_launch)  
    )  
])
```

Building and Launching the Package

With all the files in place, you can build and launch your custom ROS2 package:

1. Build the Package

```
cd /workspaces/mavlab/ros2_ws  
colcon build --packages-select gnc  
source install/setup.bash
```

2. Launch the System

```
ros2 launch gnc gnc_launch.py
```

This will start: 1. The Panisim simulator 2. The navigation node 3. The guidance control node with waypoint tracking

Configuring Waypoints

Waypoints for vessel navigation are defined in the `guidance.yaml` input file:

```
waypoints_type: XYZ  
waypoints:  
- [0.0, 0.0, 0.0]    # Starting point  
- [10.0, 0.0, 0.0]   # First waypoint  
- [10.0, 10.0, 0.0]  # Second waypoint  
- [0.0, 10.0, 0.0]   # Third waypoint  
- [0.0, 0.0, 0.0]    # Return to start
```

Monitoring System Performance

Once the system is running, you can monitor various aspects of the waypoint tracking:

```
# View odometry data
ros2 topic echo /<vessel_name>_<id>/odometry_sim

# View actuator commands
ros2 topic echo /<vessel_name>_<id>/actuator_cmd

# View waypoints
ros2 topic echo /<vessel_name>_<id>/waypoints
```

You can also visualize the vessel's movement in the web interface at <http://localhost:8000>.

Running the simulator without ROS2

Overview

Panisim provides a standalone mode that allows you to run simulations without requiring the ROS2 middleware. This lightweight mode is ideal for quick testing, development, and scenarios where the full ROS2 ecosystem is not needed.

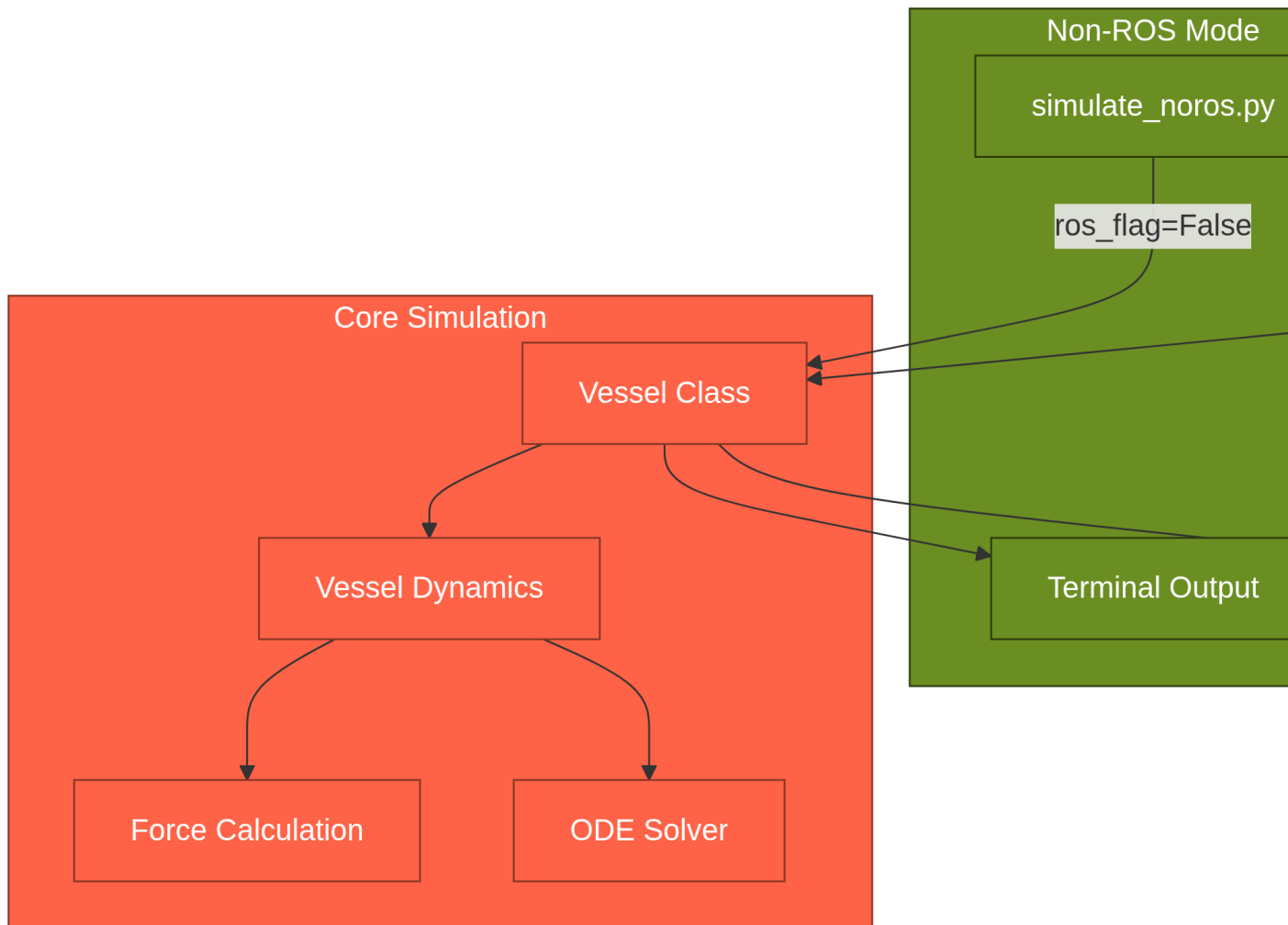
The non-ROS mode:

- Eliminates ROS2 dependencies and overhead
- Provides terminal-based feedback
- Supports the full vessel dynamics model
- Allows custom controller implementation
- Simplifies debugging of core simulation components

Implementation Details

The non-ROS mode is implemented through two key components:

1. **`simulate_noros.py`**: A standalone script that initializes and runs the simulation
2. **`class_vessel.py`**: The core vessel class with a `ros_flag` parameter that controls ROS-specific behavior



The `ros_flag` Parameter

The `ros_flag` parameter in the `Vessel` class constructor determines whether the vessel should use ROS2-specific functionality:

```
def __init__(self, vessel_params: Dict, hydrodynamic_data: Dict, vessel_id: int, ros_flag: bool):
    """Initialize vessel with parameters and hydrodynamic data.

    Args:
        vessel_params: Dictionary containing vessel parameters
        hydrodynamic_data: Dictionary containing hydrodynamic coefficients
        vessel_id: Unique identifier for the vessel
        ros_flag: Flag indicating whether to use ROS2 functionality
    """
    self.ros_flag = ros_flag
    # ... rest of initialization
```

When `ros_flag` is set to `False`:

1. No ROS2 node is created for the vessel
2. No publishers or subscribers are established
3. Command inputs must be set directly in code or through custom control logic
4. Simulation data is displayed in the terminal instead of through ROS2 topics

Running the Non-ROS Simulator

To run Panisim without ROS2:

1. Navigate to the `makara` directory
2. Run the `simulate_noros.py` script:

```
cd /workspaces/mavlab/ros2_ws/src/mav_simulator/mav_simulator
python3 simulate_noros.py
```

The simulation will use the same configuration files as the ROS2 version, loading vessel parameters, hydrodynamics, and initial conditions from the input files.

Simulation Output

The non-ROS simulator provides detailed terminal output showing:

- Simulation time
- Vessel position and orientation
- Linear and angular velocities
- Forces acting on the vessel, including:

- Hydrodynamic forces
 - Gravitational forces
 - Control surface forces
 - Thruster forces
- Control surface angles
 - Thruster RPMs

Example output:

Starting simulation...

Time [s] | Position [x, y, z] | Velocity [u, v, w]

Time: 0.00

Position [x, y, z]: [0.00, 0.00, 0.00]

Velocity [u, v, w]: [0.50, 0.00, 0.00]

Orientation [phi, theta, psi] (deg): [0.00, 0.00, 0.00]

Hydrodynamic forces: [-0.40, 0.00, 0.00, 0.00, 0.00, 0.00]

Gravitational forces: [0.00, 0.00, 0.00, 0.00, 0.00, 0.00]

Control surface forces: [0.00, 0.00, 0.00, 0.00, 0.00, 0.00]

Thruster forces: [0.00, 0.00, 0.00, 0.00, 0.00, 0.00]

Control surface angles [deg]: []

Thruster RPMs: []

Time: 0.01

Position [x, y, z]: [0.01, 0.00, 0.00]

Velocity [u, v, w]: [0.50, 0.00, 0.00]

...

Implementing Custom Controllers

One of the key advantages of the non-ROS mode is the ability to easily implement custom controllers directly in the code. The `vessel_ode` method in `class_vessel.py` contains a section specifically for non-ROS mode:

```
if not self.ros_flag:
    ## TODO: Implement your controller logic here to get the actuator commands

    ## example
    # if self.control_surface_control_type == 'fixed_rudder':
    #     self.delta_c = con.fixed_rudder(t, state, n_control_surfaces, 10.0) #Enter rudder a
```

```

# elif self.control_surface_control_type == 'switching_rudder':
#     self.delta_c = con.switching_rudder(t, state, n_control_surfaces)
# else:
#     raise ValueError(f"Invalid control surface control type: {self.control_surface_control_type}")

# # Get thruster commands
# if self.thruster_control_type == 'fixed_rpm':
#     self.n_c = con.fixed_thrust(t, state, n_thrusters, 1000.0) #Enter RPM here
# else:
#     raise ValueError(f"Invalid thruster control type: {self.thruster_control_type}")
pass

```

To implement your own controller:

1. Create a controller module (e.g., `my_controller.py`) with your control algorithms
2. Import your controller module in `simulate_noros.py`
3. Modify the `vessel_ode` method to use your controller:

```

# In vessel_ode method of class_vessel.py
if not self.ros_flag:
    # Custom controller implementation
    if hasattr(self, 'control_surface_control_type'):
        if self.control_surface_control_type == 'my_controller':
            self.delta_c = my_controller.calculate_rudder(t, state, self.n_control_surfaces)

```

Example: Adding a Simple PID Controller

Here's an example of how to add a simple PID heading controller:

1. Create a controller module:

```

# my_controller.py
import numpy as np

def pid_heading_controller(t, state, n_surfaces, desired_heading=0.0):
    """Simple PID heading controller.

    Args:
        t: Current time
        state: Vessel state vector
        n_surfaces: Number of control surfaces
    """

```

```

        desired_heading: Target heading in radians

Returns:
    numpy.ndarray: Control surface angles in radians
    """
    # Extract current heading
    current_heading = state[11] # psi is at index 11

    # Calculate heading error
    heading_error = desired_heading - current_heading

    # Normalize to -pi to pi
    heading_error = (heading_error + np.pi) % (2 * np.pi) - np.pi

    # PID gains
    Kp = 0.5
    Ki = 0.01
    Kd = 0.1

    # Calculate PID terms (simplified - would need integral and derivative state)
    p_term = Kp * heading_error

    # Calculate rudder command (assume first surface is main rudder)
    rudder_angle = np.clip(p_term, -np.pi/4, np.pi/4)

    # Create control surface command array
    delta_c = np.zeros(n_surfaces)
    if n_surfaces > 0:
        delta_c[0] = rudder_angle

    return delta_c

```

2. Modify the `vessel_ode` method in `class_vessel.py`:

```

if not self.ros_flag:
    # Import controller at the top of the file
    import my_controller

    # Use controller for rudder commands
    self.delta_c = my_controller.pid_heading_controller(
        t, state, self.n_control_surfaces, desired_heading=np.pi/4) # 45 degrees

```



```
# Set thruster commands if needed
if self.n_thrusters > 0:
    self.n_c = np.ones(self.n_thrusters) * 1000.0 # Fixed RPM
```

Customizing the Non-ROS Simulator

Modifying Simulation Parameters

You can customize the simulation by modifying the `simulate_noros.py` file:

1. Change input file path:

```
sim_params, agents = read_input("/path/to/your/custom_input.yml")
```

2. Modify output format:

```
# Customize the print statements in the simulation loop
print(f"Custom format: time={vessel.t:6.2f}, position=[{pos[0]:6.2f}, {pos[1]:6.2f}]"
```

3. Add data logging:

```
# Add at the beginning of the simulate function
import csv
log_file = open('simulation_log.csv', 'w', newline='')
csv_writer = csv.writer(log_file)
csv_writer.writerow(['Time', 'X', 'Y', 'Z', 'U', 'V', 'W', 'P', 'Q', 'R', 'Phi', 'Theta'])

# Inside the simulation loop
csv_writer.writerow([vessel.t, pos[0], pos[1], pos[2], vel[0], vel[1], vel[2],
                    vessel.current_state[3], vessel.current_state[4], vessel.current_state[5],
                    angles[0], angles[1], angles[2]])
```

Extending the Simulator

You can extend the non-ROS simulator for specific use cases:

1. Batch simulations:

```
def run_batch_simulations():
    results = []
    for initial_speed in [0.5, 1.0, 1.5, 2.0]:
        sim_params, agents = read_input("/workspaces/mavlab/inputs/simulation_input.yml")
        agents[0]['vessel_config']['initial_conditions']['start_velocity'][0] = initial_speed
```

```

        vessel = Vessel(agents[0]['vessel_config'], agents[0]['hydrodynamics'], vessel_i)
        vessel.simulate()
        # Extract results
        final_pos = vessel.history[-1, 6:9]
        results.append((initial_speed, final_pos))
    return results

```

2. Parameter studies:

```

def parameter_study():
    import matplotlib.pyplot as plt

    turning_radii = []
    rudder_angles = np.arange(-30, 31, 5) * np.pi / 180 # -30 to 30 degrees

    for rudder_angle in rudder_angles:
        # Set up simulation with fixed rudder angle
        sim_params, agents = read_input("/workspaces/mavlab/inputs/simulation_input.yml")
        vessel = Vessel(agents[0]['vessel_config'], agents[0]['hydrodynamics'], vessel_i)

        # Custom controller for fixed rudder
        vessel.control_surface_control_type = 'fixed'
        vessel.fixed_rudder_angle = rudder_angle

        # Run simulation
        vessel.simulate()

        # Calculate turning radius from final path
        # (simplified - would need actual path analysis)
        path = vessel.history[:, 6:8] # x, y positions
        turning_radius = calculate_turning_radius(path)
        turning_radii.append(turning_radius)

    # Plot results
    plt.plot(rudder_angles * 180 / np.pi, turning_radii)
    plt.xlabel('Rudder Angle (degrees)')
    plt.ylabel('Turning Radius (m)')
    plt.savefig('turning_radius_study.png')

```

Advantages of Non-ROS Mode

The non-ROS mode offers several advantages:

1. **Simplicity:** No need to set up ROS2 environment or handle ROS-specific configurations
2. **Performance:** Lower overhead without ROS middleware, especially for rapid prototyping
3. **Debugging:** Easier to debug core simulation components without ROS complexity
4. **Quick Testing:** Fast iteration for testing vessel dynamics or controller concepts
5. **Portability:** Can run anywhere Python is available, without ROS2 installation
6. **Learning:** Excellent starting point for understanding vessel dynamics before moving to ROS2
7. **Batch Processing:** Efficient for running multiple simulations for parameter studies or optimization

Limitations

Compared to the ROS2 mode, the non-ROS mode has some limitations:

1. **No Visualization:** Lacks the web-based visualization dashboard available in ROS2 mode
2. **Limited External Interaction:** No standardized way to interact with external systems
3. **No Distributed Simulation:** Cannot distribute components across multiple processes/machines
4. **No Standardized Messaging:** No access to ROS2's message types and communication patterns
5. **Manual Data Collection:** Requires custom code for data logging and analysis

Conclusion

The non-ROS mode provides a lightweight alternative for running Panisim simulations without the complexity of the ROS2 ecosystem. It's particularly useful for rapid development, testing, and educational purposes. By understanding the implementation through `ros_flag` and custom controller integration, users can leverage this mode for a wide range of applications while still benefiting from Panisim's accurate vessel dynamics modeling.